Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

North West European Shelf Production Centre
NWSHELF_ANALYSISFORECAST_PHY_LR_004_001

Issue: 3.0

Contributors: Jonathan Tinker, Robert King, Niall McConnell, Christine Pequignet, Peter Sykes, Thorger Brüning, Daniel Lea, Simon Good, Susan Kay, Matthew Martin, Andrea Rochner, Andrew Saulter, Christopher Stokes.

Approval date by the Copernicus Marine Service product quality coordination team: 24/09/2025

CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column specifies the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	28/05/2025	all	First version of document	J. Tinker, R. King, N. McConnell, C. Pequignet, P. Sykes, T. Brüning, D. Lea, S. Good, S. Kay, M. Martin, A. Rochner, A. Saulter, K. Stokes.	T. Brüning

ACRONYM TABLE

Acronym	Description
Argo	International profiling float network (www.argo.net)
CTD	CTD refers to an electronic device equipped with sensors that measure Conductivity, Temperature, and Depth
DJF	December, January February – Meteorological winter.
EAN	Estimated Accuracy Number
EiS	Enter into Service
EOF	Extended Orthogonal Function
ERSEM	European Regional Seas Ecosystem Model Biogeochemistry model (Butenschön et al., 2016).
ICES	International Council for the Exploration of the Sea
JJA	June, July August – Meteorological summer.
MAM	March, April May – Meteorological spring.
MDS	Marine Data Store
MLD	Mixed Layer Depth
MY	Multi-year
NBS	Near (sea) Bed Salinity
NBT	Near (sea) Bed Temperature
NRT	Near Real Time
NWS-IBI	The NWS-MFC system run by the IBI-MFC from 2023-2025
REP	Reprocessing
RMSD	Root Mean Square Difference
SLA	Sea Level Anomaly
SON	September, October, November – Meteorological autumn.
SSH	Sea Surface Height
SSS	Sea Surface Salinity
SST	Sea Surface Temperature
T_{4m}	Temperature at 4m depth
TAC	Copernicus Marine Service Thematic Assembly Centre
UM	Unified Model - the Met Office weather and atmospheric model
XBT	Expendable Bathythermograph

KEY DEFINITIONS

NWS shelf region (Wakelin et al., 2012).	The region mask of Wakelin et al. (2012) is used to divide the model domain into 13 regions (see Figure 3). The NRT model domain includes the North West European shelf seas, as well as other regions such as part of the North East Atlantic Ocean, the Norwegian Trench, the Skagerrak and Kattegat. When we refer to the North West Shelf region, we only refer to a portion of the model domain. The North West Shelf region is defined by combining a number of region from the Wakelin et al. (2012) region mask. These regions are: the Southern North Sea; Central North Sea; Northern North Sea; English Channel; Shetland Shelf; Irish Shelf; Irish Sea; and the Celtic Sea region.
Obsstats	A Met Office software package to assess model skill based on the model-observation output by the NEMO observations operator.

RELEVANT DOCUMENT LIST

	Ref	Title	Version
DA1	CMEMS-NWS-PUM-004-001Product User Manual for NWSHELF_ANALYSISFORECAST_PHY_LR_004_001		1
DA2	CMEMS-NWS-PUM-004-002Product User Manual for NWSHELF_ANALYSISFORECAST_PHY_LR_004_002		3
DA3	CMEMS-NWS-QUID-004-002	Quality Information Document for NWSHELF_ANALYSISFORECAST_PHY_LR_004_002	6
DA4	CMEMS-NWS-PUM-004-013	Product User Manual for NWSHELF_ANALYSISFORECAST_PHY_LR_004_013	3
DA5	CMEMS-NWS-QUID-004-013	Quality Information Document for NWSHELF_ANALYSISFORECAST_PHY_LR_004_013	3

TABLE OF CONTENTS

Acronym Table *3*

Key definitions *4*

Relevant document List *5*

I Executive summary *8*

I.1 Products covered by this document **8**

I.2 Summary of the results **9**

I.3 Estimated Accuracy Numbers **10**

II Production system description *11*

II.1 Production centres name **11**

II.2 Production system name **11**

II.3 Description of the target version **11**

II.3.1 FOAM AMM7: model components, region, grid and bathymetry 13

II.3.2 Physical model 15

II.3.3 Assimilation method 16

III Validation framework *19*

III.1 Summary of tested metrics **20**

III.2 Model feedback files and Obsstats **20**

III.3 Statistical evaluation **20**

III.3.1 Temporal and spatial summary statistics. 22

III.3.2 Regional statistics 22

III.3.3 Meteorological seasons 24

IV Validation results *26*

IV.1 Observations used for the validation **26**

IV.2 Tidal harmonics **29**

IV.2.1 Comparison with tide gauges 29

IV.3 Sea surface temperature (SST) **32**

IV.3.1 Comparison with in-situ and satellite SST data 32

IV.3.2 Comparison to OSTIA 36

IV.4 Sub-surface temperature and salinity profiles **38**

IV.4.1 Temperature and salinity profiles 39

IV.4.2 Surface and bed temperature and salinity from ICES temperature and salinity profiles 40

IV.4.3 Glider 43

IV.5 Mixed Layer Depth **49**

IV.5.1 Comparison with Gliders 49

IV.6 Sea Level Anomaly **50**

IV.6.1 Satellite sea level anomalies 50

V System's Noticeable events, outages or changes ***53***

VI Quality changes since previous version ***54***

VII References ***55***

VIII Appendix ***58***

VIII.1 Additional Figures **58**

VIII.2 Additional Tables **62**

I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document refers to the following real-time CMEMS products:

- NWSHELF_ANALYSISFORECAST_PHY_LR_004_001. The physical component of the North West European Analysis and Forecast system (~7 km resolution).

Products at higher temporal and spatial resolution (1.5 km) are available for this region, in a complementary product known as NWSHELF_ANALYSISFORECAST_PHY_004_013.

Table 1: List of NWS-NRT variables available in the NWSHELF_ANALYSISFORECAST_PHY_LR_004_001 product.

Variable	Vertical resolution	Horizontal resolution	Temporal resolution
Temperature	All levels, surface to bottom	7 km	daily average
Salinity	All levels, surface to bottom	7 km	daily average
Zonal velocity	All levels, surface to bottom	7 km	daily average
Meridional velocity	All levels, surface to bottom	7 km	daily average
Sea surface elevation	Surface	7 km	daily average
Mixed Layer Depth	N/A	7 km	daily average
Bottom temperature	Bottom	7 km	daily average

I.2 Summary of the results

The quality of the NWS lower resolution near real time (NRT) physical ocean forecast system (based on the FOAM-AMM7-ERSEM model; e.g. Butenschön et al., 2016; Edwards et al., 2012; Fowler et al., 2023; Skákala et al., 2018; Tonani et al., 2019) has been assessed using two-year trial experiments (years 2022-2023). has been assessed using two-year trial experiments (years 2022-2023).

The FOAM-AMM7-ERSEM model is forced using atmospheric flux fields from the Met Office global Unified Model (UM) (Willett et al., 2025; Xavier et al., 2024) and boundary conditions from the Met Office Global 1/12° operational forecasting system (ORCA12; Barbosa Aguiar et al., 2024). FOAM-AMM7 operates with two daily analysis cycles, in which observations are assimilated to update the model state, followed by a 7-day forecast. We have evaluated the FOAM-AMM7 extensively against a wide range of variables and observations. Here we summarise the performance against each. extensively against a wide range of variables and observations. Here we summarise the performance against each.

Tides: The tides are accurate across most of the domain. When comparing the modelled and observed M_2 tide (the dominate (semi-diurnal) tidal constituent) at tide gauges, the median relative amplitude bias is -3.03% (-3.4 cm) and the overall Root Mean Squared Difference (RMSD) amplitude is 12.19 cm – this is small compared to the median M_2 amplitude of 127.29 cm. Figure 2), where the tides are very small.

Sea Surface Temperature (SST): SST data are assimilated by this system, so surface differences are very small with a mean (cold) bias typically of about -0.05°C (RMSD of ~0.5°C), with the exact values depending on which observation the data set is compared to. There is typically a cooler bias in summer, and coastal areas may have a warm bias.

Sub-surface temperature[1]: The mainly off shelf temperature profiles suggest a slight sub-surface warm bias (of 0.09°C, with an RMSD of 0.47°C). Near-bed temperature estimates from the ICES (International Council for the Exploration of the Sea) data set suggest AMM7 has a small on-shelf cold bias -0.06°C with a large RMSD (1.31°C), reflecting a warm bias under summer stratification in the North Sea.

Salinity[1]: The mainly off shelf salinity profiles (i.e., from beyond the continental shelf) suggest a near-zero salinity bias (of ~0.00 psu, with an RMSD of 0.09 psu). Surface salinity estimates from the ICES data set suggest AMM7 has a small on-shelf median bias (0.04 psu) but with a large RMSD (3.80 psu), reflecting a large model-too-salty bias downstream of the Baltic outflow. The ICES near-bed salinity statistics are much better, as the Baltic outflow is a predominately surface feature.

Sea Level Anomaly (SLA): AMM7 SLA has a small bias and RMSD (-0.56 cm and 7.13 cm respectively), with a positive bias around Scotland and the Southern North Sea offset by a larger negative bias in the open ocean. The RMSDs are larger along the European coast and in the Irish Sea.

[1] The NWS subsurface is relatively sparsely observed, and so assessing the model sub-surface skill (temperature, salinity and the mixed layer depth) is limited by observation availability. We do not draw on the conclusions from the glider analysis for this summary section due to their localised coverage, therefore we don't include MLD.

I.3 Estimated Accuracy Numbers

Estimated Accuracy Numbers (EAN) are summary statistics of the model "best estimate" analysis performance (see Figure 1). Overall, the model is relatively unbiased, and so 95% confidence intervals might be reasonably estimated using 1.96×RMSD (see III.3 for a more formal calculation of confidence intervals from the RMSD).

Table 2: Estimated accuracy numbers for the NWSHELF_ANALYSISFORECAST_PHY_LR_004_001 analysis performance over the years 2022-2023.

Variable	Location	Supporting observations	Mean Difference (observation-model)	RMS Difference
M_2 tidal harmonic (amplitude)	Full domain	Tide gauge data	-3.03% (-3.4 cm)	12.19 cm
M_2 tidal harmonic (phase)	Full domain	Tide gauge data	-11.2 minutes (-5.41°)	11.49°
SST	Full domain	*In situ* observations Obsstats	-0.05°C	0.42°C
	Full domain	Satellite observations Obsstats	-0.03°C	0.40°C
	Full domain	OSTIA satellite L4	-0.07 °C	0.33 °C
Sub-surface Temperature	Full domain	*In situ* profile observations	0.09°C	0.47°C
	NWS	ICES NBT	-0.06°C	1.32°C
	Summer North Sea	ICES NBT	0.66°C	-
Salinity	Full domain	*In situ* observations	~0.00	0.07 PSU
	NWS	ICES SSS	0.04psu (median bias)	3.80psu
Sea level anomaly	Full domain	Altimeter from satellite	-0.56 cm	7.13 cm
Mixed layer depth	limited areas	Gliders	Relative bias ~15%	Relative RMSD <50%

II PRODUCTION SYSTEM DESCRIPTION

This modelling system (hereinafter the NWS-MFC system) is from a new NWS-MFC consortium and so is different to the previous system (hereinafter the NWS-IBI system). It is similar to the system available prior to 2023, which was also delivered by the UK Met Office (hereinafter the Met Office system). See Appendix Table 6 for the dates of these changes.

II.1 Production centres name

North-West Shelf MFC -Met Office, UK.

II.2 Production system name

North-West European Shelf Physics Analysis and Forecast (NWSHELF_ANALYSISFORECAST_PHY_LR_004_001).

Met Office – FOAM (Forecasting Ocean Assimilation) Shelf Seas Atlantic Margin Model 7 km (AMM7)

II.3 Description of the target version

This document details the quality of the physical ocean products from the FOAM Shelf Seas –Atlantic Margin Model at 7 km horizontal resolution (FOAM-AMM7). These data are intended for use with biogeochemistry data generated from the model's ERSEM component (European Regional Seas Ecosystem Model; Butenschön et al., 2016). FOAM-AMM7 has a higher resolution companion model with a coupled wave model, running at 1.5-km (FOAM-AMM15), which is assessed in a separate QUID [DA5]. Table 3 presents a summary of the differences, with fuller details in the Appendix (Appendix Table 1, Appendix Table 2, Appendix Table 3, Appendix Table 4 and Appendix Table 5), and in the associate PUMs ([DA1] and [DA4]). This system includes an online Biogeochemistry model, ERSEM, which produces the NRT BGC analysis product (NWSHELF_ANALYSISFORECAST_BGC_004_002, [DA2], [DA3]). It is a one-way coupled system, which means that the BGC does not provide feedback to the physical model.

Table 3: Overview of the AMM7 NWSHELF_ANALYSISFORECAST_PHY_LR_004_001 system, with summary differences from the AMM15 NWSHELF_ANALYSISFORECAST_PHY_004_013 system. This table summarises Appendix Table 1 - Appendix Table 5 in the Appendix.

NWSHELF_ANALYSISFORECAST_		**AMM7v9 PHY_LR_004_001**	**AMM15 PHY_004_013**
Model	Bathymetry	GEBCO corrected by NOOS partners	EMODnet 2015[2]
	Horizontal resolution	7 km	1.5 km
	Model Grid 40°N- 65°N 20°W – 13°E Regular grid	40°N-65°N 20°W-13°E Regular grid	~ 45°N - 63°N ~ 20°W - 13°E The grid has a rotated pole, chosen so that the grid-equator runs through the domain to reduce the distortion of cells with increasing latitude. While the rotated latitude is constant, the longitudinal grid steps range from ~1.47 km to 1.5 km.
Forcings	Surface forcing	Met Office Global Unified Model (MetUM) Atmospheric model NWP analysis and forecast fields, applied as direct fluxes.	
	Lateral boundaries	Met Office global ocean 12km model ORCA12, daily mean 3D Temperature, Salinity and barotropic currents, and sea surface height	
		Daily river climatology	Daily river climatology
Data Assimilation	NEMOVAR version	V3	V4
	Data Assimilated	SST (*in situ*, and satellite), temperature and salinity profiles, glider data, sea level anomaly data, ocean colour chlorophyll	SST (*in situ*, and satellite), temperature and salinity profiles, glider data, sea level anomaly data

The FOAM-AMM7 system delivers a daily updating 2-day analysis (Best Estimate and NRT analysis) and 7-day forecast (Figure 1). The previous 2 years of Best Estimate analysis available are from the Marine Data Store. The 2-day analysis cycle provides a large assimilation window, and therefore allows more data to be assimilated.

[2] Bathymetry was obtained from the EMODnet Portal: EMODnet Bathymetry Consortium, EMODnet Digital Bathymetry (DTM), EMODnet Bathymetry (September 2015 release)

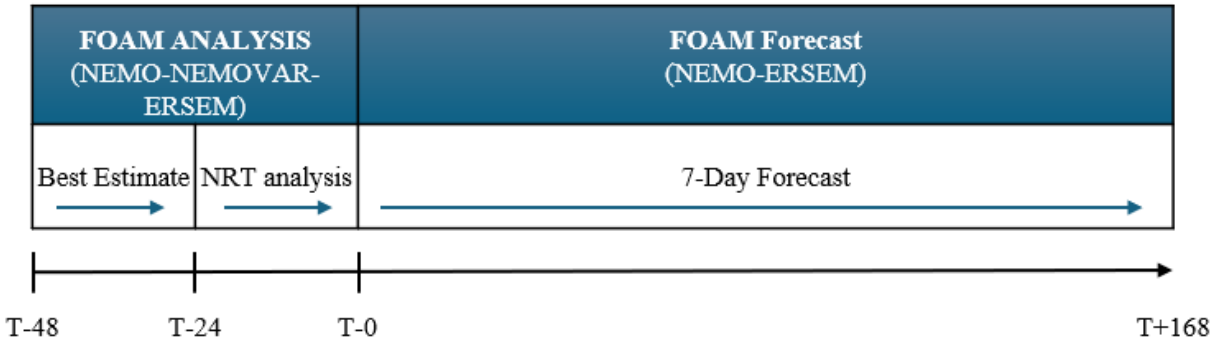


Figure 1: FOAM-AMM7 production cycle, including the two analysis cycles (the "Best Estimate", and the "NRT Analysis"), and the 7-day forecast.

II.3.1 FOAM AMM7: model components, region, grid and bathymetry

The Forecasting Ocean Assimilation Model (FOAM) 7 km Atlantic Margin model (AMM7) is a hydrodynamic model, (one-way) nested within the Met Office global ocean model, with lateral boundary data taken from the CMEMS Baltic model. The hydrodynamics are supplied by the Nucleus for European Modelling of the Ocean (NEMO) with the NEMOVAR 3D-Var FGAT system used for the assimilation of Sea Surface Temperature, Sea Level Anomaly, and temperature and salinity profiles data (see Table 5).

AMM7 is (one-way) coupled to the biogeochemistry model ERSEM, which is the basis of the NRT BGC analysis product (NWSHELF_ANALYSISFORECAST_BGC_004_002). See Appendix Table 5, the PUM [DA2], and CMEMS-NWS-QUID-004-002 [DA3], for details of ERSEM.

The model is implemented on the European North-West continental Shelf (NWS), from 40°N, 20°W to 65°N, 13°E, on a regular lat-lon grid with 1/15° latitudinal resolution and 1/9° longitudinal resolution (approximately 7 km square). The domain extends beyond the shelf to include some of the adjacent North-East Atlantic, but the focus of this system is on the shelf itself and the deep water is primarily included to ensure there is appropriate cross-shelf exchange. The FOAM AMM7 domain is shown in Figure 2. The CMEMS products delivered from this system cover the full model domain (i.e. on and off shelf).

It uses a hybrid s-sigma terrain following coordinate system (Siddorn and Furner, 2013) with 51 levels, employed to retain vertical resolution on the shelf. To reduce horizontal pressure gradient errors over extreme topography, the scheme includes a z-S hybrid as described in Madec et al. (1996). The loss of vertical resolution at these points is more than compensated for by reduced errors in the horizontal pressure gradient term. A key bathymetric feature separating the shelf from the deep ocean is the shelf slope, which runs south to north from Portugal to Norway. The combination of very steep bathymetry and sigma coordinates requires special treatment for modelling horizontal pressure gradients, which is done using a Pressure Jacobian formulation. Bathymetry was supplied by North-West Shelf Operational Oceanographic System (NOOS) partners, who have processed GEBCO 1 arc-minute data together with a variety of other local data sources. The bathymetry was further interpolated in-house to fit with the model grid.

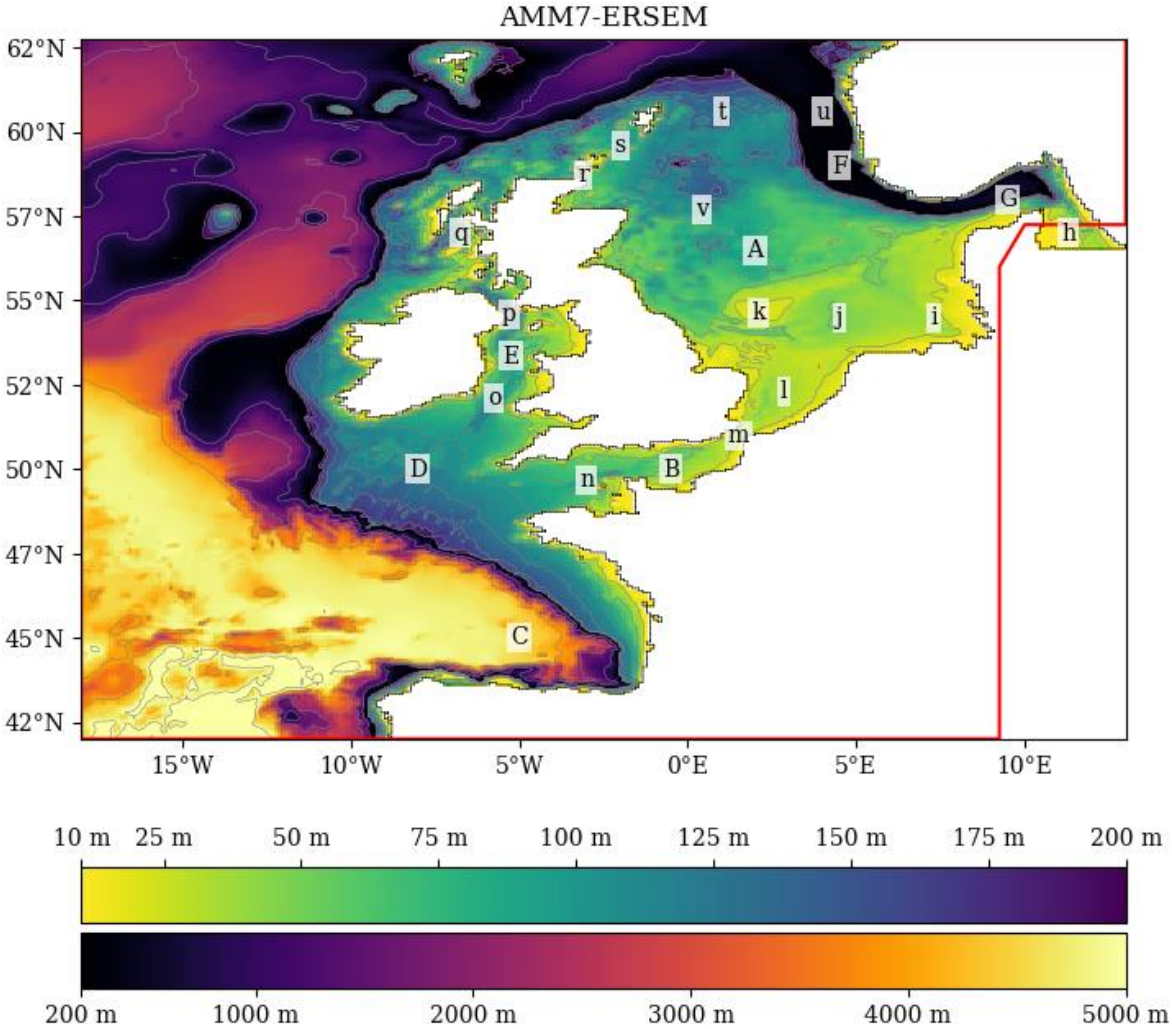


Figure 2: FOAM AMM7 bathymetry (m)). The red box is the domain covered by the AMM7 product delivered to CMEMS. Letters indicate major (upper case) and minor (lower case) features which are given in Table 4.

Table 4: Major (upper case) and minor (lower case) features within the North West Shelf region as given in Figure 2, with the plotted locations.

	NWS Feature name	Location
A)	North Sea	56.50°N 2.00°E
B)	English Channel	50.00°N 0.50°W
C)	Bay of Biscay	45.00°N 5.00°W
D)	Celtic Sea	50.00°N 8.00°W
E)	Irish Sea	53.35°N 5.25°W
F)	Norwegian Trench	59.00°N 4.50°E
G)	Skagerrak	58.00°N 9.50°E
h)	Kattegat	57.00°N 11.30°E
i)	German Bight	54.50°N 7.30°E
j)	Oyster Ground	54.50°N 4.50°E
k)	Dogger Bank	54.64°N 2.09°E
l)	Southern Bight	52.30°N 2.85°E
m)	Dover Strait	51.00°N 1.50°E
n)	Hurd's Deep	49.67°N 3.00°W
o)	St George's Channel	52.10°N 5.75°W
p)	North Channel	54.60°N 5.30°W
q)	Sea of the Hebrides	57.00°N 6.75°W
r)	Pentland Firth	58.73°N 3.10°W
s)	Fair Isle Channel	59.60°N 2.00°W
t)	Northern North Sea Inflow	60.60°N 1.00°E
u)	North Sea Outflow	60.60°N 4.00°E
v)	Dooley Current	57.70°N 0.40°E

II.3.2 Physical model

The FOAM system is based on version 3.6 of the Nucleus for European Modelling of the Ocean (NEMO) code (Madec et al., 2017). This is a community ocean model that has a wide user and developer base particularly in Europe. Tidal forcing is included both on the open boundary conditions via a Flather radiation boundary condition (Flather, 1976) and through the inclusion of the equilibrium tide. The external elevation and depth mean velocity was determined from 15 tidal constituents taken from a tidal model of the northeast Atlantic (Flather, 1981). The model is one-way nested within the Met Office operational ORCA12 global ocean model (1/12°) and the Copernicus operational Baltic Sea model. They

provide temperature, salinity, sea surface height (not at the Baltic boundary), and depth integrated currents at the open boundaries.

Freshwater river fluxes are from a climatology of daily discharge data for 279 rivers from the Global River Discharge Data Base (Vörösmarty et al., 2000) and from data prepared by the Centre for Ecology and Hydrology as used by Young and Holt (2007). Surface forcing is provided from the Met Office's Numerical Weather Prediction model using three-hourly heat and moisture fluxes, and hourly instantaneous fields of wind speed and surface pressure. The 1 band shortwave radiation light attenuation scheme previously used in POLCOMS (the Proudman Oceanographic Laboratory Coastal-Ocean Modelling System – see Holt and James, 2001) is used to calculate the light attenuation profile in the water column.

Tidal modelling requires a non-linear free surface and this is facilitated in NEMO by using a variable volume layer method. The short time scales associated with tidal propagation and the free surface require a time splitting approach, splitting modes into barotropic and baroclinic components. The bottom boundary condition includes a log layer representation and a k-epsilon turbulence scheme is implemented with a generic length scale(Umlauf and Burchard, 2003). The model uses a non-linear free surface, an energy and entropy conserving form of the momentum advection, and a free slip lateral momentum boundary condition. The tracer equations use a TVD advection scheme(Zalesak, 1979). For tracer diffusion a Laplacian diffusion scheme on geopotential surfaces is applied, whereas for momentum diffusion a mixed Laplacian/bilaplacian scheme is used, with the Laplacian operator applied on geopotential surfaces and the bilaplacian on model surfaces.

II.3.3 Assimilation method

For the North West Shelf, assimilation of Sea Surface Temperature (SST), Sea Level Anomaly (SLA) and temperature and salinity (T/S) profiles is performed using a 3DVar first-guess-at-appropriate-time (FGAT) algorithm implemented using the NEMOVAR software (see

Appendix Table 4). Operationally, assimilation proceeds in three steps. First, a one-day model forecast is performed, during which observations are read in and model values are interpolated to the observation locations at the nearest model time step. The observation and model information, together with seasonally-varying information about the errors in the observations and model background. are then used by NEMOVAR to generate 3D increments of temperature, salinity and velocity as well as sea surface height. A multivariate balancing scheme, based on physically-derived relationships, is used to ensure the resulting increments are dynamically balanced. Finally, another run of NEMO is carried out where the increments are added onto the model fields using an incremental analysis update scheme(IAU, Bloom et al., 1996). In the operational system, these 3 steps are carried out twice, the first step from T-48 to T-24, and the second step from T-24 to T0, and a forecast is then run. The analysis at T-24 is used on the next day's operational run as the initial conditions at the start of the T-48 analysis window.

In-situ data is obtained through various sources, including the CMEMS in-situ TAC and the Global Telecommunications System (GTS[3]). Swath satellite SST data (L2P data) are obtained from GHRSST[4]. At present this includes geostationary infrared SST data from the SEVIRI instrument, polar orbiting infrared data from the AVHRR, VIIRS and SLSTR instruments, and polar orbiting microwave data from the AMSR2 instrument. All data are quality controlled using the method of Ingleby and Huddleston(2007). The satellite altimeter data are obtained from CMEMS SL-TAC[5] along with the corrections necessary for use with a wind- and pressure-forced, tidal coastal model. These data, along with the model background, are detided (i.e., tide signal removed) before the increments are calculated. Additionally, the NEMOVAR assimilation system uses a variational bias correction scheme, the implementation of which is detailed by Barbosa Aguiar et al.(2024). This bias correction scheme aims to remove time varying biases in SST data by correcting data to a reference data set of assumed unbiased SST observations (While and Martin, 2019). An on-line bias correction scheme is used to reduce errors in the Mean Dynamic Topography (used in the assimilation of satellite altimeter SLA data). This scheme is based on Lea et al. (2008), and the bias is estimated using time-averaged corrections to the sea surface height produced during the assimilation process.

The observations used for the assimilation are presented on Table 5.

[3] http://www.wmo.int/pages/prog/www/TEM/GTS/index_en.html
[4] www.ghrsst.org
[5] EUROPEAN OCEAN ALONG-TRACK SEA LEVEL ANOMALIES NRT TAILORED FOR DATA ASSIMILATION, product name: SEALEVEL_EUR_PHY_ASSIM_L3_NRT_OBSERVATIONS_008_043

Table 5: List of observations used for data assimilation.

Type	Fields	Platforms/Satellite	Data source
IN SITU	SST Temperature and salinity profiles	• Ships • Drifters • Fixed moored arrays • Gliders • XBTs • CTDs • ARGO • Ferry boxes • Recopesca buoys* • Thermosalinograph	GTS; http://www.wmo.int/pages/prog/www/TEM/GTS CMEMS-INS-TAC : http://marine.copernicus.eu/ INSITU_GLO_NRT_OBSERVATIONS_013_030
SATELLITE	SLA Along Track* **along with the corrections necessary for the use with a wind and pressure forced, tidal coastal model. Previously, SLA was only assimilated in water of depths >700m*	• Cryosat • Altika • Jason 3 • Sentinel 3 • Sentinel 6	CMEMS SL TAC: http://marine.copernicus.eu/ SEALEVEL_EUR_PHY_ASSIM_L3_NRT_OBSERVATIONS_008_043
	SST L2p/L3c	• NOAA-AVHRR • MetOp-AVHRR • SEVIRI • VIIRSG • AMSR2	Group for High-Resolution Sea Surface Temperature (GHRSST) www.ghrsst.org

As detailed in King et al.(2018), the NEMOVAR system includes bias correction schemes for both SST and for altimeter data (using the CNES-CLS09 Mean Dynamic Topography of Rio et al. (2011) as a reference). The SLA bias correction is unchanged in AMM7. The AMM7 SST bias correction is improved in AMM7. A variational data correction scheme is added to the observations-of-bias scheme to improve the SST bias correction. This new system is more resilient to changes in the observing system.

III VALIDATION FRAMEWORK

Period of trial: 2 years, 1 January 2022 to 31 December 2023

Systems: We base our assessment on a pair of FOAM-AMM7 and FOAM-AMM15 simulations outlined below (both run with NEMOVAR data assimilation). These are the same setups used operationally, with full details provided above, and in Appendix Table 1 - Appendix Table 5, and in the PUM [DA1].

Simulations: The assessment simulations are part of a wider suite of trials run at the Met Office to assess individual components of the system. Here we only focus on the simulations that will be delivered to Copernicus Marine. These simulations have been extended to the present day, and have been delivered as the 2-year back catalogue.

Table 6: Evaluation simulations overview.

Simulation's name:	**AMM7, AMM15**
Data assimilated	3D SST, SLA, Sub-surface profiles, ChlA for ERSEM.
Initial state	Taken from the Operational System.
Surface forcings	Met Office Global Coupled-NWP (direct fluxes).
Lateral boundary conditions	Met Office ORCA12 model.
Simulation period	1/1/2022-31/12/2023.

III.1 Summary of tested metrics

Table 7: List of tested metrics.

Variable	Description	Supporting observations	Class	Metrics
Tidal harmonics	European coasts	Tide gauges data	2	Amplitude and phase: scatterplots, co-tidal charts, median bias (model-observation), relative bias, and RMSD.
SST	Shelf, whole region	In-situ (buoys and ships of opportunity)	1 and 4	Bias (model-observation), RMSD
	Shelf, whole region	Satellite SST	1 and 4	Bias (model-observation), RMSD
	whole region	OSTIA L4 SST analysis	1	Bias (model-observation), RMSD
Temperature and salinity profiles	whole domain; sub regions	Profile data available over GTS; glider data not assimilated;	4	Bias (model-observation), RMSD.
Surface and near bed temperature and salinity	Shelf	ICES profile dataset	4	Bias (model-observation),
Mixed layer depth	sub regions	glider data assimilated	4	Bias (Observation-model); RMSD, time series

III.2 Model feedback files and Obsstats

As part of the assimilation system, NEMO runs a daily cycle without assimilation, where it reads in the observations, and interpolates the model value to the location and time of the observations. It then produces a feedback file which pairs the observations with the interpolated model equivalent value. This is then used by the NEMOVAR data assimilation scheme to calculate the model increments. These feedback files provide a useful estimate of model's skill at simulating the observations, prior to their assimilation. The "Obsstats" package uses this feedback files to assess the model skill, by first binning the model-minus-observation anomalies into daily 2°x2° bins (larger bins for the sparce temperature and salinity profile data). By taking the sum, and the sum of the square of the anomalies, with the number of observations, the bias and RMSD can be easily calculated, for the full data set, or as spatial or temporal means. We use the Obsstats approach throughout this assessment.

III.3 Statistical evaluation

The basis of our evaluation is a comparison between our model and the observations for "best estimate" (Figure 1, day minus 2 analysis, from T-48-T-24h). While observations have known (and unknown) errors,

for the purpose of this evaluation, we treat the observations as the truth and test the model's differences from what is observed. We therefore use model minus observations anomalies, where a positive anomaly indicates that the model is greater than the observation (e.g. a warm anomaly suggests the model is too warm, also known as a warm bias).

We rely on bias

$$Bias = \frac{\sum x}{N} \tag{1}$$

and root mean square difference (RMSD)

$$RMSD = \sqrt{\frac{\sum(x^2)}{N}} \tag{2}$$

to summarise the model performance, where *x* is the model minus observation anomalies, and *n* is the number of observations. If we assume observations represent the true state of the ocean, the bias and RMSD can be interpreted as measures of the model's accuracy and precision. The bias shows, on average, how close the model is to the observed truth, and the RMSD measures the range of deviation of the model from the observed values.

For some variables, the bias and RMSD is easier to interpret in the context of the typical values. For example, the tide amplitude bias varies widely across constituents, so it more meaningful to consider the bias in the context of the mean amplitude. We therefore also compute the relative bias and RMSD by normalising (dividing) the bias and RMSE by the mean values of the model and observations - 0.5*(model + observation).

For a normally distribution, 95% of the data is within 1.96 standard deviations (σ) of the mean, and so ±1.96σ give 95% confidence intervals. The variance (σ^2) is the square of the standard deviation is calculated as

$$\sigma^2 = \frac{\sum(x-\bar{x})^2}{N} = \frac{\sum(x)^2}{N} - \left(\frac{\sum x}{N}\right)^2 \tag{3}$$

Which can be calculated from the bias and RMSD,

$$\sigma^2 = (RMSD)^2 - (Bias)^2 \tag{4}$$

As can the σ

$$\sigma = \sqrt{(RMSD)^2 - (Bias)^2} \tag{5}$$

As the bias approaches 0, σ tends to the RMSD.

$$\lim_{Bias \to 0} \sigma = RMSD \tag{6}$$

Therefore, when the model is relatively unbiased, and the 95% confidence intervals might be reasonably estimated using 1.96×RMSD.

III.3.1 Temporal and spatial summary statistics.

The model and observations are defined in both space and time, and depending on the context, we sometimes provide timeseries of the daily bias and RMSD (calculated from all the observations for a given day), typically presenting the mean of the time series. In other places, we show the bias and RMSD for all anomalies within a grid box (or region) and then plot maps of the summary statistics. The spatial mean of these maps is not directly comparable to the means of the time series. Finally, we also tabulate the bias and RMSD of all the anomalies within a region (or full domain) over the whole period giving a statistic that is not directly comparable to either of the first two approaches. In the following text, we clearly indicate which approach is used. The difference between these approaches tends to be bigger for the RMSD than for the bias, and is more important with irregular sparce observations (such as *in situ* SST), than for gridded data (such as satellite or OSTIA SST).

III.3.2 Regional statistics

In the following report, we often give regional summaries of the results. We base these on the region mask of Wakelin et al. (2012), which is presented in Figure 3a. For some applications, particularly when comparing to sparse data, we need larger regions and therefore, we have combined the Wakelin regions into larger regions: the Validation regions (Figure 3b). When referring to the NWS, we use the NWS mask (Figure 3c) which is also based on the Wakelin region mask.

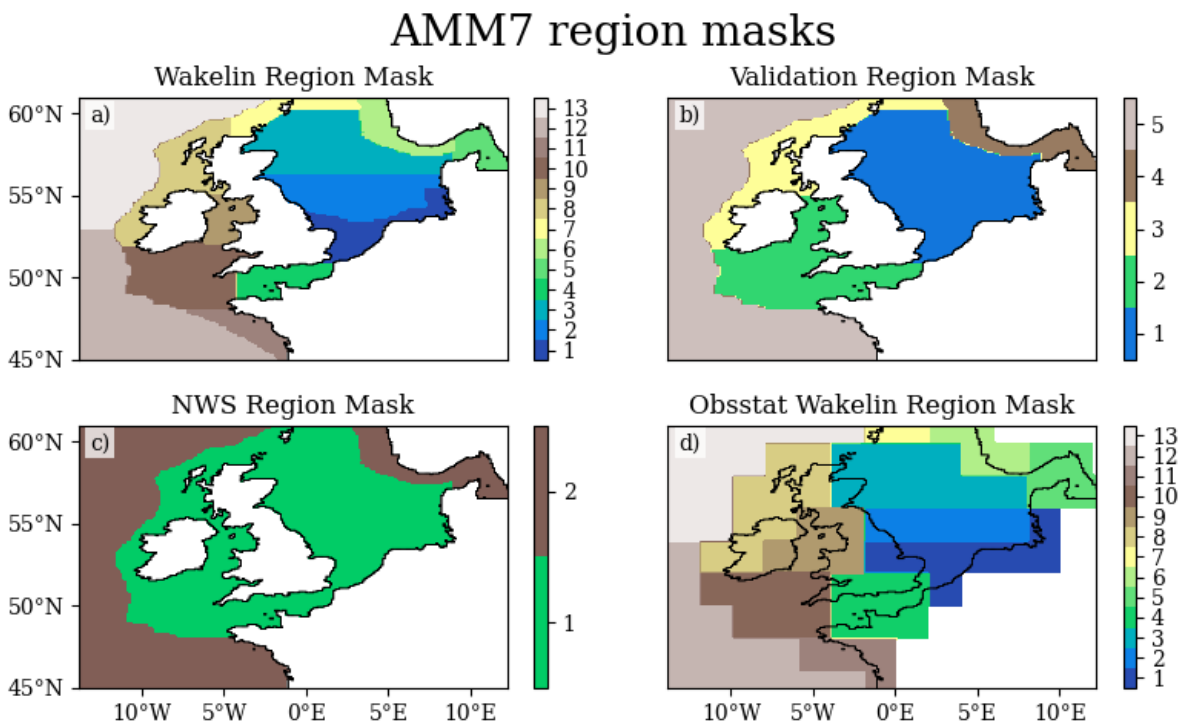


Figure 3: Region masks used in this evaluation. a) the Wakelin Region mask, based on Wakelin et al. (2012), b) the Validation mask (i.e., Wakelin regions joined together to give larger regions for comparison with sparse observations), c) the NWS mask (NWS delineated from the ocean, and the Norwegian Trench/Skagerrak and Kattegat also based on the underlying Wakelin regions), d) the Obsstats region mask, which is the Wakelin region mask on a 2°x2° grid. Region names given in Table 8.

The Obsstats bins the observation into 2°x2° bins prior to analysis, and so the Wakelin mask is resampled to this grid (by taking the median regions within each grid box) in Figure 3d. The region names are given in Table 8.

Table 8: Names of regions with Region masks in Figure 3

Mask number	Mask region	Region Name	Based on concatenation of Wakelin regions numbers
Wakelin Region Mask/Obsstat Wakelin Region Mask			
1, 4	1	Southern North Sea	
1, 4	e	Central North Sea	
1, 4	3	Northern North Sea	
1, 4	4	English Channel	
1, 4	5	Skagerrak/Kattegat	
1, 4	6	Norwegian Trench	
1, 4	7	Shetland Shelf	
1, 4	8	Irish Shelf	
1, 4	9	Irish Sea	
1, 4	10	Celtic Sea	
1, 4	11	Armorican Shelf	
1, 4	12	NE Atlantic (South)	
1, 4	13	NE Atlantic (North)	
Validation Region Mask			
2	1	North Sea	1, 2, 3.
2	2	Celtic Seas	4, 9, 10.
2	3	Outer Shelves	7, 8.
2	4	Norwegian Trench /Skagerrak/Kattegat	5, 6.
2	5	NE Atlantic	11, 12, 13.
NWS Region Mask			
3	1	NWS (North -West Shelf)	1, 2, 3, 4, 7, 8, 9, 10.
3	2		5, 6, 11, 12, 13.

III.3.3 Meteorological seasons

We often use seasonal means to display the data. We use meteorological seasons, where December to February (inclusive) is meteorological winter, March to May is spring, June to August is summer and September to November is autumn. These are hereinafter, DJF, MAM, JJA, SON respectively. As our trials are 2 years, running from January 2022 to December 2023 they start and finish near the middle of winter. We average the first two months of the trial as the first winter (January and February of 2022, hereinafter 2022 JF), and the last month of the trial (December 2023) as the last winter (hereinafter 2023 D). Between these two partial seasons, we have 7 full seasons (hereinafter 2022 MAM, 2022 JJA, 2022 SON (in 2022),

22/23 DJF for the 2022/2023 winter, and 2023 MAM, 2023 JJA and 2023 SON). This gives 9 (partial and complete) seasons, which are used as the basis of *Figure 13*, *Figure 14*, *Figure 15*, Appendix Figure 1, *Appendix Figure 3*, *Appendix Figure 4*.

IV VALIDATION RESULTS

The validation is based on the operational system described section II.2, running the simulations outlined in section III. Validation presented is based on the simulation period in Table 6. These simulations have been extended to the November 2025 Enter into Service (EiS), and have been delivered as the 2-year back catalogue.

We focus on the bias, and RMSD to quantify the differences between the models and the observations. We use model minus observation biases, where a warm bias indicates the model is warm relative to the observations. When summarising the bias or RMSD (i.e. when a time series of the bias or RMSD is given), we typically provide the mean value, but occasionally use the median when the data doesn't appear to be normally distributed.

IV.1 Observations used for the validation

The observations used for the validation of the experiments are:

- Set of observations (*in situ*, see Figure 4, and satellite, Figure 5) used by the data assimilation, see Table 5. These measurements are considered semi-independent because even if the misfit (observation-model) is computed before the model is corrected by the data assimilation, there is a dependency between model-observations due to the previous data assimilation cycles. SST is also evaluated against the VIIRSG satellite product.

The Independent observations, not used by data assimilation, are:

- Met Office Operational Glider transect at 59°N (from Orkneys (2.5°W) to 0°);
- COSYNA HF radar in the German Bight;
- Copernicus INS-TAC Tide gauges (INSITU_GLO_PHY_SSH_DISCRETE_MY_013_053);
- ICES Temperature and Salinity Profiles (High Resolution CTD data from the ocean hydrochemistry dataset[6] extracting profiles for 2022 and 2023 between 46°N-62.75°N and 16°W-10°E);
- Copernicus SST-TAC OSTIA SST analysis[7].

[6] https://data.ices.dk/view-map?theme=201809
[7] SST_GLO_SST_L4_NRT_OBSERVATIONS_010_011

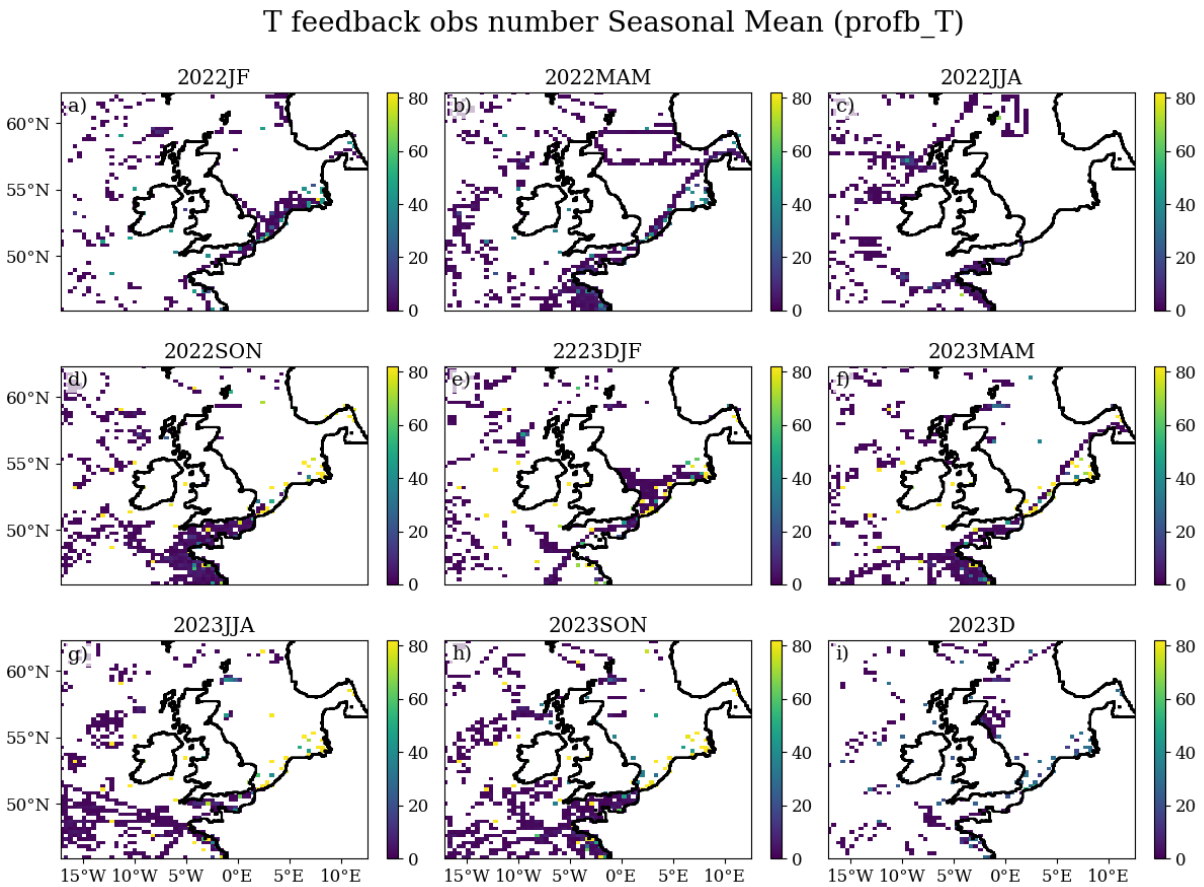


Figure 4: Total number of temperature profiles in four grid box bins for 2022-2023, presented by season. Meteorological seasons are used (MAM (March-May), JJA (June-August), SON (September-November) and DJF (December- February) for spring, summer, autumn and winter respectively), while 2022JF and 2023D refer to January-February 2022 and December 2023 respectively. See Appendix Figure 1 for the equivalent salinity profile plot.

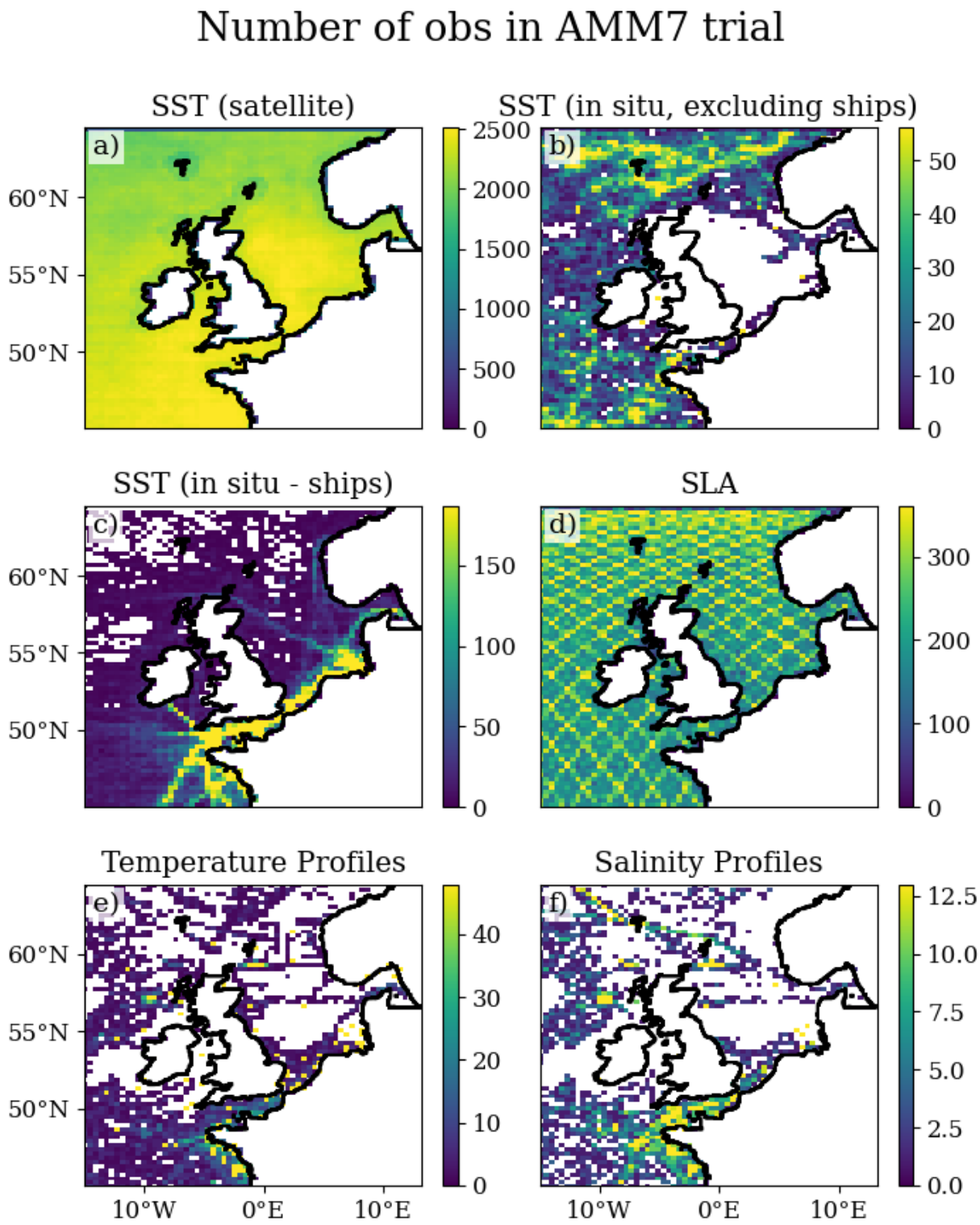


Figure 5: Number of observations (2022-2023) used by the data assimilation and by the validation. a) satellite SST b) in situ SST based on drifters and moored buoys c) in-situ SST based on ship of opportunity observations, d) satellite SLA, e) temperature profiles, f) salinity profiles.

IV.2 Tidal harmonics

AMM7 simulates the tides well. The dominant M_2 component has a small amplitude bias of typically ~-3.03% of the tidal amplitude (3.4 cm, with a RMSD of 12.19 cm, Table 9). The model has a phase offset for most tide gauges, leading to a M_2 phase difference of 11.2 minutes (model -5.41° too late, with a RMSD of 24 mins). In most areas, the tide is well represented, although the model skill is lower in the Skagerrak and Kattegat.

IV.2.1 Comparison with tide gauges

A harmonic analysis of the dominant tidal constituents was undertaken on the model and tide gauge observations from the Copernicus INS TAC product (INSITU_GLO_PHY_SSH_DISCRETE_MY_013_053), using a common 2-year period (2022-2023), to supplements the tidal evaluation against a 20-year simulation undertaken during model development(Graham et al., 2018). Tide gauges were excluded from the assessment if their time series contained less than 80% data within the two-year period, if the tide gauge was located more than two grid boxes inland from the AMM7 coast, or if they were considered not to be on the open coast (e.g., in estuaries, fjords, or the Wadden Sea). This left 146 tide gauges for comparison. Appendix Figure 2 classifies the tide gauges as being used in the analysis, or why they were excluded. A least squares tidal fit was performed on 7 constituents (M_2, S_2, N_2, O_1, K_1, Q_1, M_4) selected in accordance with the Rayleigh Criterion (Foreman, 1977). The summary statistics for these tidal constituents, based on all included tide gauges is included in Table 9. Positive phase biases suggest the modelled tide is too early.

Table 9: Summary statistics of the tidal harmonic analysis, based on 146 tide gauges from the Copernicus INS TAC tide gauge product included in the analysis.

	Period (h)	Median Amp ½(model+ observations; cm)	Median Amp Bias (model-observations; cm)	Median relative Amp Bias (%)	Amp RMSD (cm)	Median Phase bias (°)	Median Phase bias (min)	Phase RMSD (°)	Phase RMSD (min)
M_2	12.42	-3.40	127.29	-3.03	12.19	-5.41	-11.20	11.49	23.78
S_2	12.00	-0.13	40.14	-0.91	3.69	-6.46	-12.93	14.00	28.00
N_2	12.66	-0.58	22.59	-2.42	2.01	-5.98	-12.62	10.67	22.52
K_1	23.93	0.84	8.17	9.30	1.18	2.73	10.87	13.79	55.03
O_1	25.82	-0.94	9.09	-11.35	1.20	-0.42	-1.82	30.94	133.14
M_4	6.21	0.09	9.20	1.09	4.62	-40.73	-42.16	61.75	63.91
Q_1	26.87	0.44	4.07	11.28	0.58	-17.41	-77.95	20.37	91.20

As an indicator of performance, we focus on the M_2 tidal constituent , which is the dominant tidal component of the NWS – M_2 is a primary lunar semi-diurnal tide with a 12.42 hour period (approximately twice daily).. When looking across all 146 tide gauges, the median relative difference in the M_2 amplitude is -3.03% (model -3.4cm too small, with a RMSD of 12.19 cm, Table 9). The median M_2 phase difference is 11.2 minutes (model -5.41° too late), with a RMSD of ~24 min (11.49°). For S_2 (principal solar semidiurnal tide), the median amplitude difference is -0.91% (-0.13 cm, RMSD = 3.69 cm) and the median phase difference of -12.93 minutes (-6.46°, RMSD = 28.0 mins, 14.0°). These relative statistics are summarised in Table 9, while additional statistics (including percentile ranges) are given in Appendix Table 7.

Figure 6 compares the model against observations for both the M_2 amplitude (cm) and phase (degrees) and shows an excellent agreement between the model and tide gauges, and a high correlation for amplitude (r = 0.99). There is an offset between the modelled phases against observed ones, which impacts the correlation (r = 0.66).

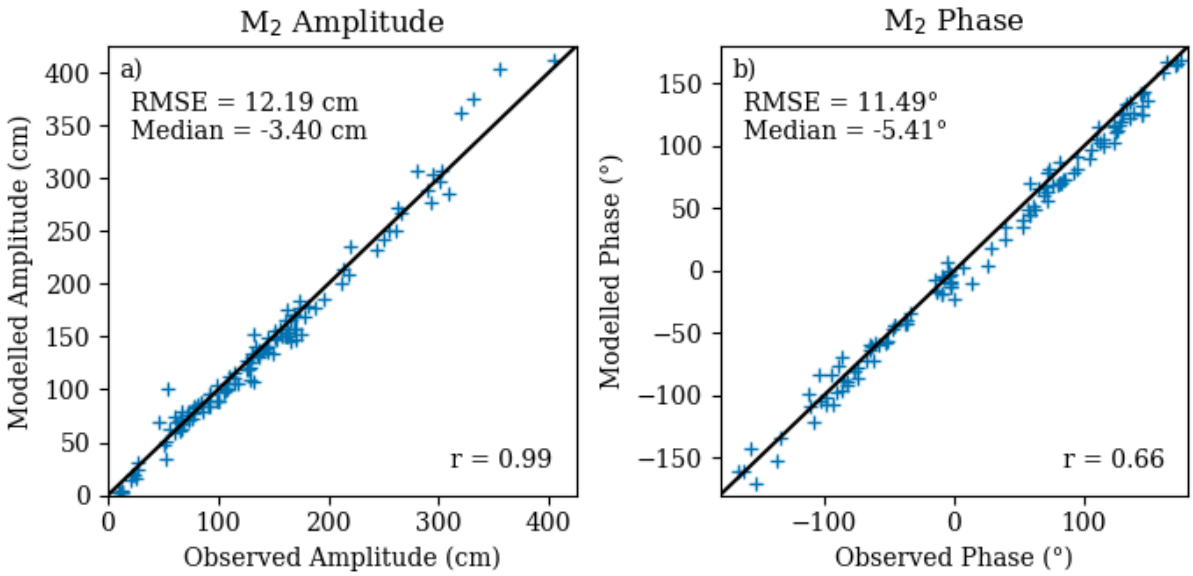


Figure 6: M_2 amplitude (a) and phase (b) observations versus model values.

Figure 7 shows the spatial distribution of the M_2 tidal errors in AMM7. The M_2 amplitude tends to be generally underestimated across the domain, however, most of the UK (apart from northern Wales), west of Normandy, and west of Jutland have overestimated M_2 amplitudes. The Skagerrak has low tidal ranges, so even a small absolute underestimation of the M_2 amplitude results in a very large relative error (Figure 7b).

As noted in Figure 6, there is a generally underestimation of the phase across the domain, which is greatest along the southern coast of the North Sea. This region is relatively close to the amphidromic point centred at ~55°N, 5°E, and so the phase changes quickly over a relatively short distance. Furthermore, it has a complex coastline with shallow waters which are unlikely to be well represented with the 10 m minimum depth implemented in AMM7. This means that AMM7 is unlikely to model the correct wave speed of the tides, which would impact the modelled phases.

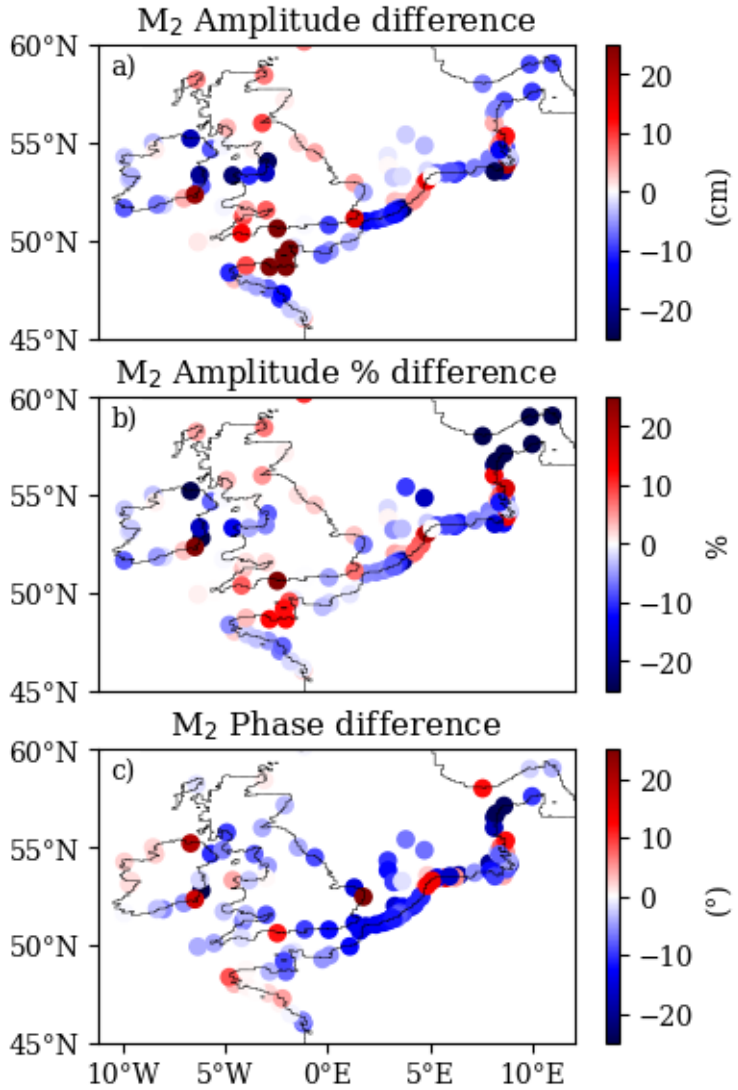


Figure 7: AMM7 M_2 tidal characteristic compared to the Copernicus tide gauges (model minus observation). a) M_2 amplitude difference, b) M_2 amplitude relative percentage difference; c) M_2 phase difference.

The co-tidal chart for M_2 (Figure 8) compares very well with observations (see comparable chart in Howarth and Pugh, 1983). The differences between AMM15 and AMM7 systems are very small (co-tidal map of AMM15 not shown).

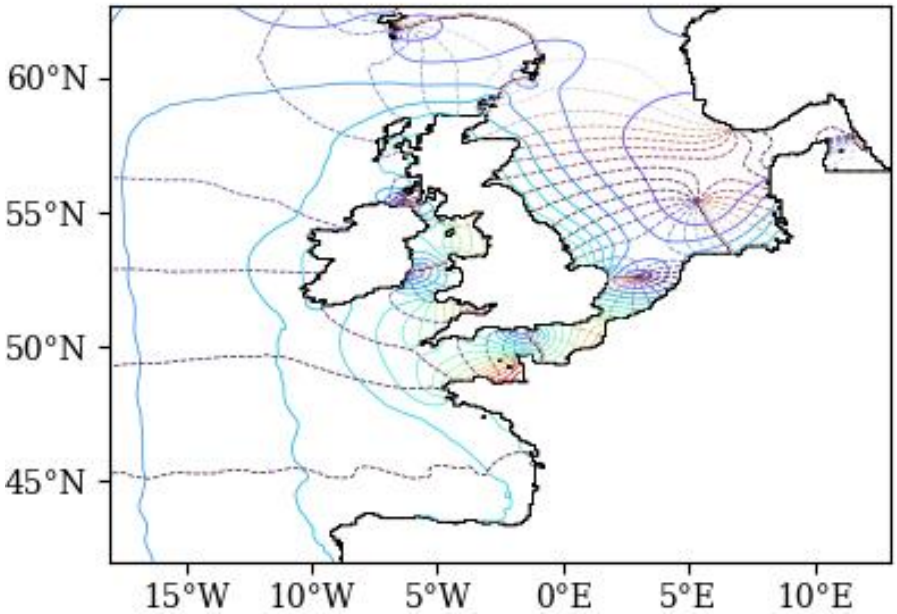


Figure 8: Co-tidal chart for M_2 component, AMM7.

IV.3 Sea surface temperature (SST)

The AMM7 model is very good at simulating SST, reflecting the amount of observed SST data assimilated. Here we assess AMM7 SST against *in situ* and satellite observation using the Obsstats package (III.2) and against the OSTIA SST analysis, drawing similar conclusion from all three methods. We show that AMM7 has a very small (cold) bias, of the order -0.05°C, and an RMSD of ~0.5°C. There is a seasonal cycle to the bias, with typically cooler biases in the summer, and a suggestion that coastal areas may have a warm bias.

IV.3.1 Comparison with in-situ and satellite SST data

SST data is assimilated in the AMM7 model (Figure 5) during the Incremental Analysis Update (IAU) step of each model run. As result, an assessment of the model's skill at predicting SST compared to observations would be expected to produce a positive result. Model SST has been assessed here against in-situ SST measurements using the Met Office's Obsstat package (as model minus observations). This package uses the model to observation matchups created before the SST data has been assimilated, and so quantify the skill of the "free-running" model. The matchups are created by interpolating the model field to the observation location at the time of the observations. The in-situ measurements are from buoys and ship of opportunity in Figure 9 and satellites in Figure 10. Figure 9 (and Figure 10). We consider the daily SST model minus observations in terms of bias, RMSD, number of observations, as maps, and as time series of domain mean statistics (Figure 9 and Figure 10). The upper rows of Figure 9 and Figure 10 show maps of the mean bias (a), RMSD averaged over the full 2-year trial (b), the total number of observations in each grid box (c). The lower rows (d) show a time series of bias (blue) and RMSD (red) as averaged over the model domain (with the temporal mean), while the lower right panels e) gives the number of observations each day. The mean daily RMSD is 0.409°C with a cold bias of -0.064°C. The cold bias is mainly due to the spring and summer months when the model is slightly cooler than the observations. The same statistics on-shelf shows very similar results.

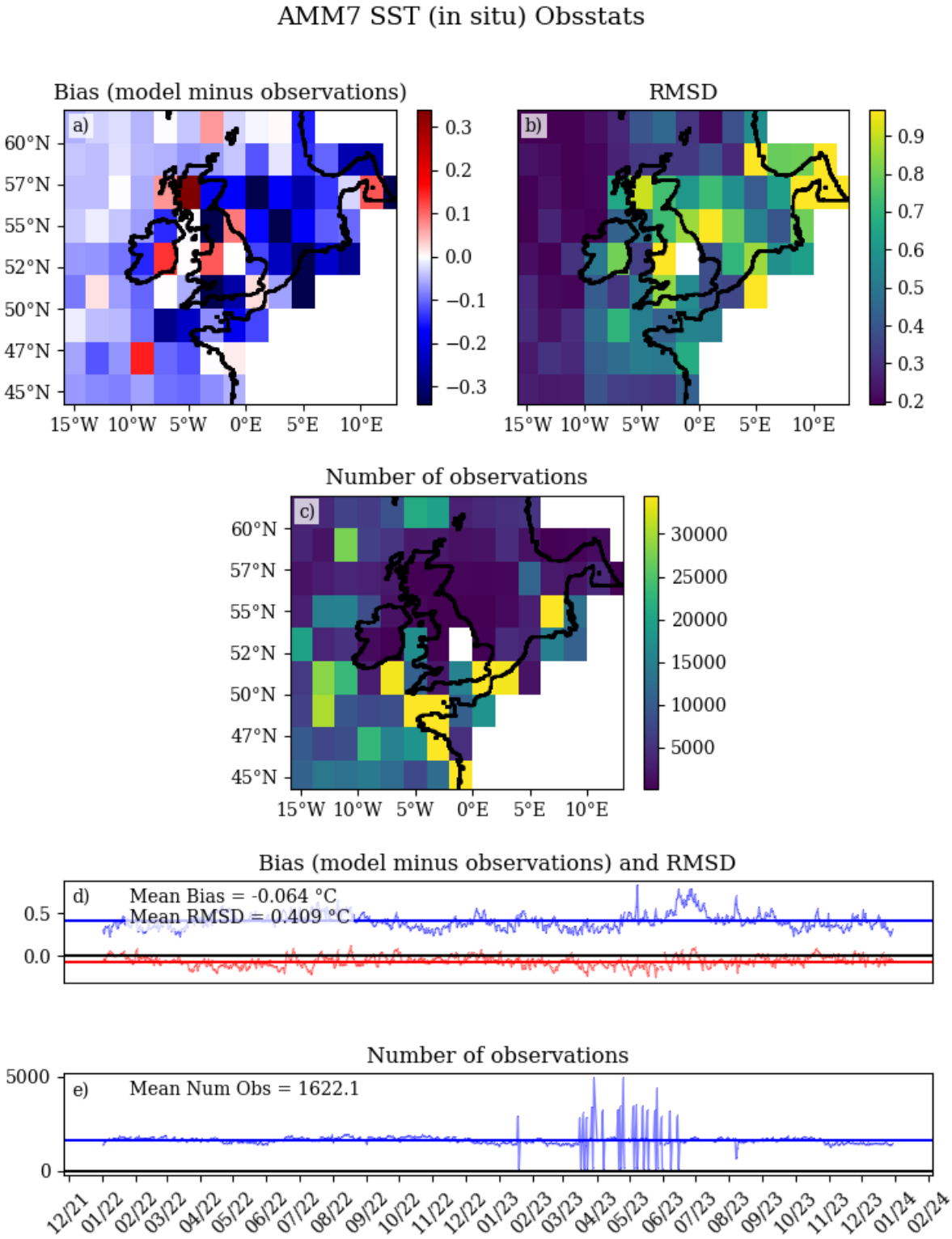


Figure 9: Model minus Observation daily SST assessment against in-situ observations. Maps of Bias, RMSD and Number of observations for 2022-2023 (a, b, c respectively) and time series of d) domain mean bias (blue) and RMSD (red) and e) number of observations per day.

Table 10: SST (in situ) regional mean Obsstats summaries, calculated from all observations within the region, as defined in Figure 3d.

Region	Bias (°C)	RMSD (°C)	Number of Observations	Number of 2°x2° bins
NWS	-0.09	0.46	372924	35
Full Domain	-0.05	0.42	992178	94
Southern North Sea	0.01	0.57	65497	3
Central North Sea	-0.12	0.43	43597	5
Northern North Sea	-0.16	0.43	25442	9
English Channel	-0.08	0.45	43044	1
Skagerrak Kattegat	0.11	1.15	3936	1
Norwegian Trench	-0.03	0.58	6387	2
Shetland Shelf	-0.02	0.26	7127	2
Irish Shelf	-0.05	0.33	27074	7
Irish Sea	0.00	0.35	16837	1
Celtic Sea	-0.13	0.45	144306	7
Armorican Shelf	-0.02	0.49	137454	2
NE Atlantic S	-0.04	0.30	304161	29
NE Atlantic N	-0.02	0.28	167316	25

When using the same analysis with satellite data, we see similar results, albeit the greater number of observations give more confidence and nuance to the interpretation. The SST bias is near zero when averaged over the domain (bias = -0.028°C). The largest RMSDs are in the coastal regions. Note that many of the coastal grid boxes are largely on land, so they have only few data points (see Figure 10c), and therefore have little impact on the domain mean time series in Figure 10d and e.

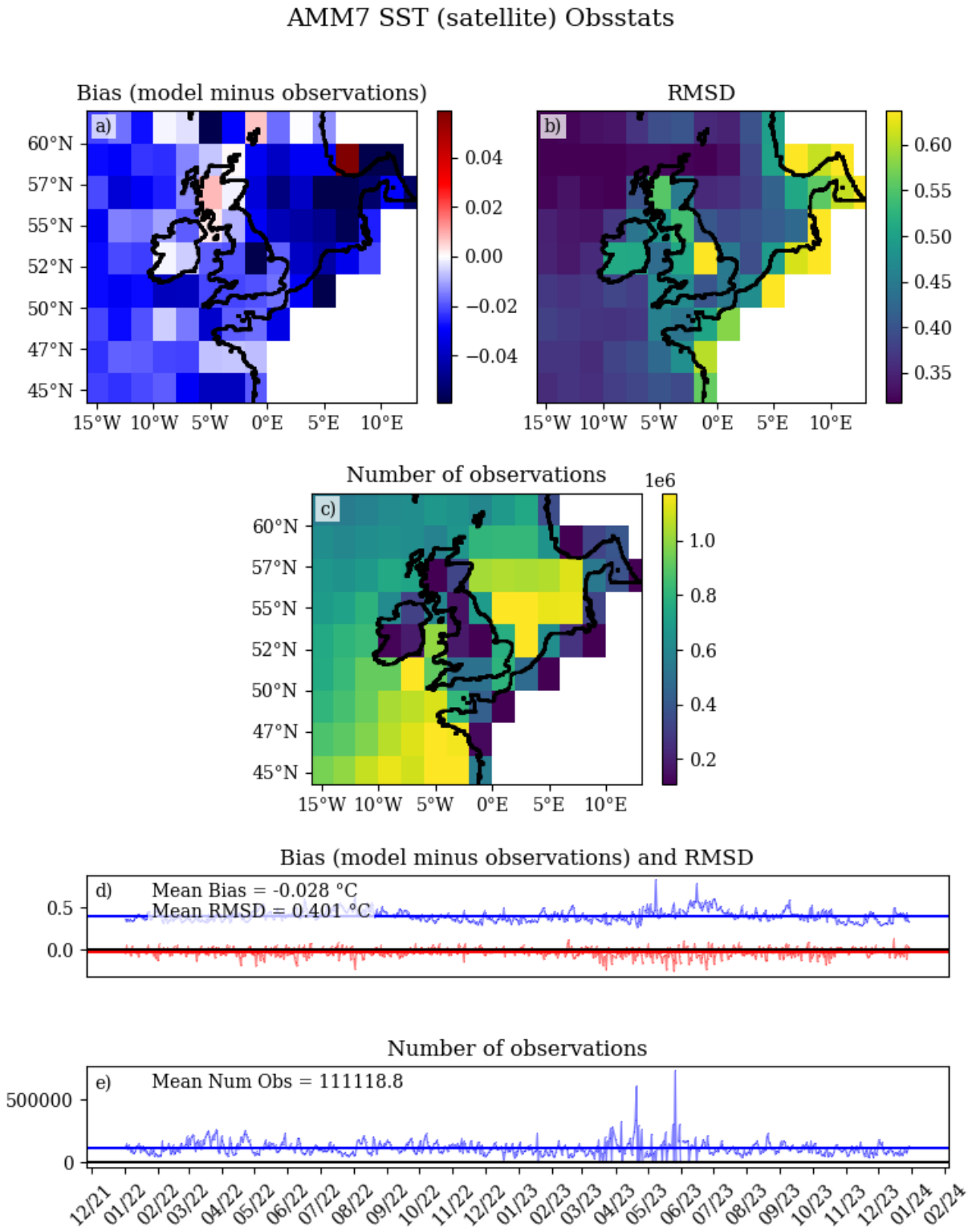


Figure 10: Model minus Observation daily SST assessment against satellite observations. Maps of Bias, RMSD and Number of observations for 2022-2023 (a, b, c respectively) and time series of d) domain mean bias (blue) and RMSD (red) and e) number of observations per day.

Table 11 SST (satellite) regional mean Obsstats summaries, calculated from all observations within the region, as defined in Figure 3d.

Region	Bias (°C)	RMSD (°C)	Number of Observations	Number of 2°x2° bins
NWS	-0.03	0.41	31587757	35
Full Domain	-0.03	0.40	73328724	94
Southern North Sea	-0.03	0.49	2841154	3
Central North Sea	-0.05	0.41	5385926	5
Northern North Sea	-0.04	0.39	8393109	9
English Channel	-0.02	0.46	810772	1
Skagerrak Kattegat	-0.03	0.60	373379	1
Norwegian Trench	-0.01	0.47	1394633	2
Shetland Shelf	-0.01	0.35	1283916	2
Irish Shelf	-0.01	0.37	4717508	7
Irish Sea	-0.02	0.47	1001678	1
Celtic Sea	-0.03	0.40	7153694	7
Armorican Shelf	-0.01	0.48	2382043	2
NE Atlantic S	-0.03	0.37	23464488	29
NE Atlantic N	-0.03	0.34	14126424	25

IV.3.2 Comparison to OSTIA

In addition to the in-situ observation assessment, the models SSTs have been compared to the Met Office's Operational Sea Surface Temperature and sea Ice Analysis (OSTIA) system (Good et al., 2020) . OSTIA provides analyses of the foundation SST (i.e. the SST free of diurnal variability) and assimilates in-situ and satellite observations. The OSTIA data are available through the CMEMS catalogue as product SST_GLO_SST_L4_NRT_OBSERVATIONS_010_011. OSTIA is produced on a 1/20°grid (~6 km resolution), however, this is not the feature resolution of the product, which depends on other aspects of the system such as the correlation length scales used in producing the analysis. OSTIA has a maximum feature resolution of ~20-30 km and so both AMM7 and especially AMM15 are expected to represent smaller features than OSTIA. We compare OSTIA to the modelled temperature linearly interpolated to 4m below the surface to represent the Foundation temperature used in OSTIA.

We have provided a time series of the spatial mean bias (model minus observations), and RMSD, when averaged over the Wakelin et al. (2012) NWS shelf region (Figure 11), finding a mean bias of -0.067°C, suggesting the model is slightly too cool, with a RMSD of 0.299°C. These statistics are reported for Wakelin et al. (2012) regions in Table 12 and the appendix, and spatial maps of the bias and RMSD broken down by season are presented in the appendix (*Appendix Figure 3*and *Appendix Figure 4*).

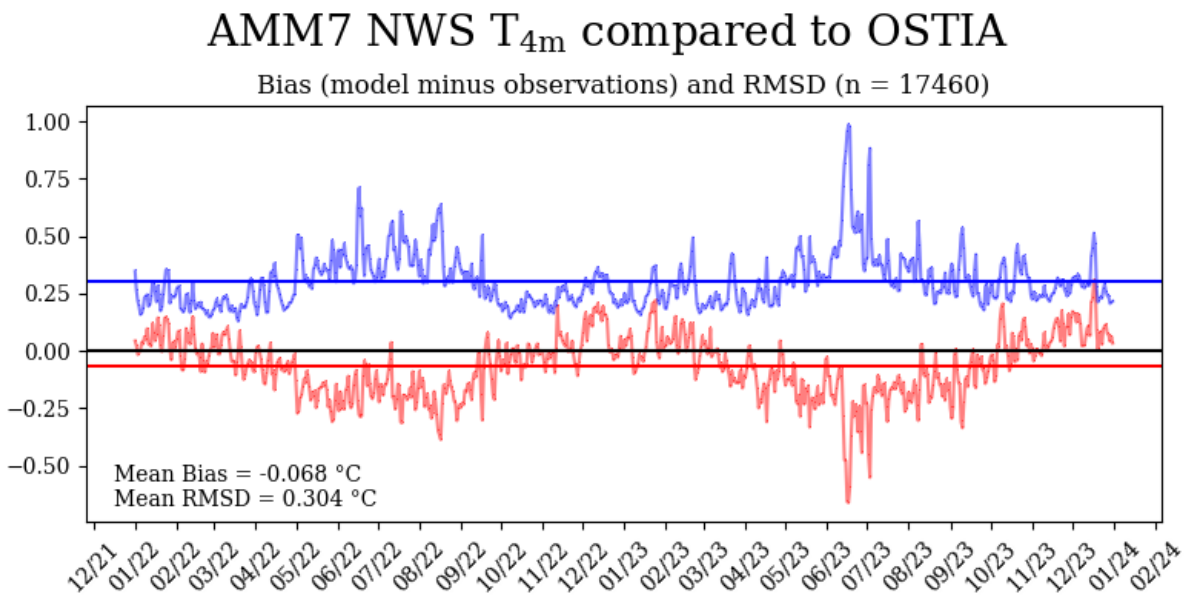


Figure 11: Time-series statistics of model T4m compared to OSTIA foundation SST. Daily bias and RMSD are calculated from all data within the model domain each day. Mean Bias and RMSD are the average value of the time series, thus they are not directly comparable to those calculated for the Wakelin regions (see Table 12).

Statistics from all model–observation anomalies within a given region (Table 12) indicate a cool bias of −0.07 °C and an RMSD of 0.33 °C for the full domain. (reported in the EAN table, Table 2). We also give seasonal maps of the bias and RMSD in the appendix (*Appendix Figure 3*and *Appendix Figure 4*).

There is a clear seasonal cycle in the SST biases, with the summer month being cooler than the winter months (when compared to OSTIA). When looking at *Appendix Figure 3*, we see that the winter months are warmer than OSTIA around the coast, and cooler than OSTIA in deeper waters, which are still closer to OSTIA than in the summer. The winter warmer coastal areas tend to cancel out the otherwise general cool bias, and so explain the smaller winter SST biases. This is consistent with what we found in section IV.3.

Table 12: Summary statistics for the evaluation of T_{4m} against OSTIA.

Region	Bias (model minus observation; °C)	RMSD (°C)	Region size (number of grid boxes)
NW European Shelf	-0.07	0.33	12710880
Full Domain	-0.07	0.33	48730136
Southern North Sea	-0.06	0.38	1249248
Central North Sea	-0.10	0.33	2165072
Northern North Sea	-0.09	0.33	3001544
English Channel	-0.03	0.34	764400
Skagerrak Kattegat	-0.07	0.66	529984
Norwegian Trench	-0.04	0.55	944944
Shetland Shelf	-0.05	0.26	931112
Irish Shelf	-0.04	0.30	1765400
Irish Sea	-0.01	0.33	586040
Celtic Sea	-0.08	0.31	2248064
Armorican Shelf	-0.06	0.40	777504
NE Atlantic S	-0.09	0.29	15324400
NE Atlantic N	-0.05	0.33	18442424

IV.4 Sub-surface temperature and salinity profiles

There are relatively few sub-surface observations on the NWS, making it difficult to evaluate the model's sub-surface and water-column structure. We use two main sources – comparing the Obsstats of the profiles assimilated into the model (prior to assimilation), which are largely off-shelf, and limited to those that are available in real time, and the ICES high resolution CTD profile dataset which focus on the NWS.

The assimilated profiles suggest that the model has a small sub-surface warm bias (of 0.09°C, with an RMSD of 0.47°C), and no salinity bias (0.00 psu, with an RMSD of 0.07psu) when averaged over depth from the mainly oceanic profiles.

The ICES SSTs tend to agree with the AMM7 having a cold bias, although the bias and RMSD are larger than with the other SST data sets. SSS are dominated by very large anomalies downstream of the Baltic outflow, and so are not normally distributed. The median bias is small (0.04psu), but the RMSD (3.80 psu) and mean bias (1.01 psu) reflect this area of larger salinity anomalies. Care must be taken when using the modelled salinity in the Skagerrak, Kattegat, and even Norwegian Trench. The bottom salinities have much smaller anomalies. NBTs, when averaged over the domain and year, have a small bias -0.06°C with a large RMSD (1.31°C) which reflects a warm bias under summer stratification in the North Sea (e.g. the 2022 summer -JJA- is 0.66°C).

The AMM7 model performs very well when compared with observations from one of the gliders, and well with those from the other one, with cold surface mixed layer temperature biases of -0.08°C and -0.3 °C (RMSE 0.18°C and 0.72°C) respectively. Sub mixed layer temperature biases are also very small in the one glider, but higher in the other, with a cold bias of -1.09°C (RMSD 1.51°C) – these are dominated by the biases nearer to shore. The salinity (fresh) biases are small (~0 and ~0.15 psu), as are the RMSDs (~0.15

psu). Due to the relatively small area of the domain assessed by the gliders, we do not include them in the key summary.

IV.4.1 Temperature and salinity profiles

We use the same approach as for the in-situ temperature to assess the model's performance in simulating temperature and salinity profiles. Obsstats are calculated from T&S profiles prior to assimilation, so quantify “free running” model performance. Profiles available in real time (and suitable for assimilation) are overwhelmingly in the open ocean part of the domain (~30 of the ~600 profiles/day are from the on-shelf region, e.g. Figure 4), and so we concentrate on the statistics for this region. AMM7 has a mean warm bias of 0.09°C, with an RMSD of 0.47°C), when averaged over the profile (Figure 12a). The salinity profiles have a near zero salty bias of 0.00 psu, with an RMSD of 0.07 psu (Figure 12b)

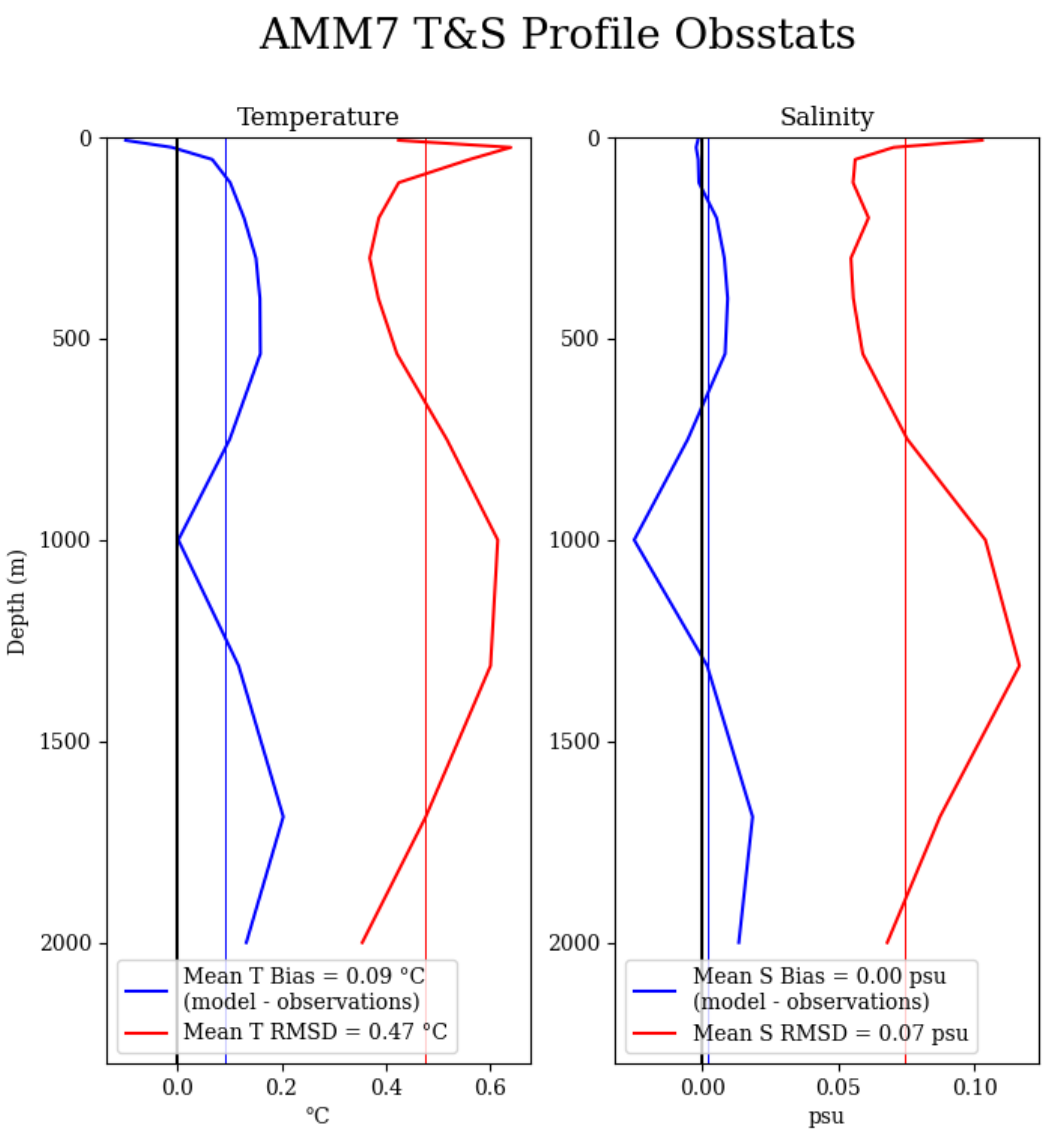


Figure 12: Temperature (left) and salinity (right) profile assessment for the full domain. The mean (model-observation) bias (red) and RMSD (blue) are given (thick lines) – the respective depth mean values are given with thin lines. These profiles are largely from the off-shelf region, as less than 10% of the profiles are from the on-shelf region).

IV.4.2 Surface and bed temperature and salinity from ICES temperature and salinity profiles

We have used the ICES (International Council for the Exploration of the Sea) temperature and salinity profiles dataset to help evaluate the near bed temperature and salinity. The ICES temperature profile data set is a delayed-mode product, and so provides more comprehensive coverage than the data available in near real time. We have extracted profiles within the model domain for 2022-2023, and matched them to the nearest model grid box. The profiles are then linearly interpolated onto the model depth levels, to allow a direct comparison. The upper value, if it is within 10 metres of the surface, is considered sea surface value. The lowest value, if it is within 10 metres of the seabed, and where the water depth is greater than 30 metres, is considered to be the near-bed value – these thresholds have been selected to ensure that the near-bed value is in the nearbed mixed layer. The model minus observation biases are then summarised in Table 13, and SST, SSS and NBT plotted in Figure 13, Figure 14 and Figure 15 respectively. Further details of the seasonal cycle for the North Sea, are available in Appendix Table 8 in the appendix).
The mean SST bias is -0.11°C, with an RMSD of 0.61°C, which is higher than that seen in other SST analyses, but the sign of the bias is consistent with the other analyses. The NBTs have a relatively small mean (cold) bias of -0.06°C (median of -0.11°C). Figure 15 shows that summer NBT in the south eastern North Sea are warm, which is quantified in Appendix Table 8.

The salinity bias has a median value of 0.04 psu suggests that most observations agree with model, but model is too salty at the Baltic outflow, and downstream into the Norwegian Trench. The large biases in this small region dominate the mean (1.01 psu, RMSD = 3.80 psu), and the difference in the mean and the median suggest the data are not normally distributed. The NBS tend to be smaller than the SSS, reflecting the large positive SSS biases associated with the Baltic outflow and Norwegian Trench being largely a surface phenomenon.

Table 13: Summary of ICES statistics for the full model domain.

	Mean (std dev)	**Median (10th-90th percentile)**	**RMSD**	**Number of observations**
SST	-0.11°C (0.61°C)	-0.13°C (-0.59°C-0.39°C)	0.61°C	3336
SSS	1.01psu (3.67psu)	0.04psu (-0.50psu-2.50psu)	3.80psu	3256
NBT	0.06°C (1.31°C)	-0.11°C (-0.69°C-1.07°C)	1.32°C	2092
NBS	-0.11psu (0.46psu)	-0.04psu (-0.31psu-0.09psu)	0.48psu	2092

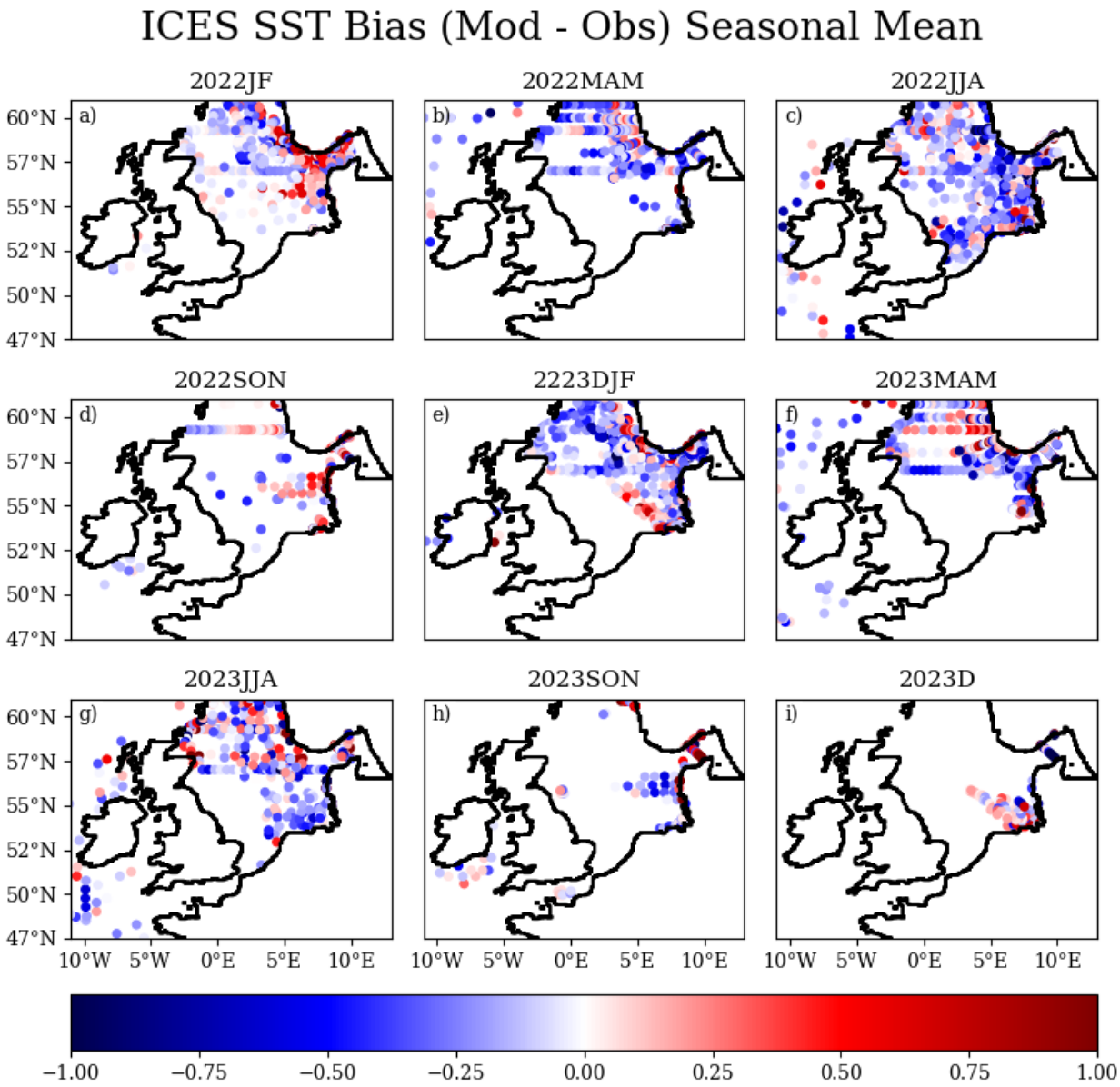


Figure 13: Seasonal model minus observation anomalies relative to ICES SST. DJF, MAM, JJA and SON refer to winter, spring, summer and autumn. See Meterological seasons (section 0) for further clarification of the season definitions.

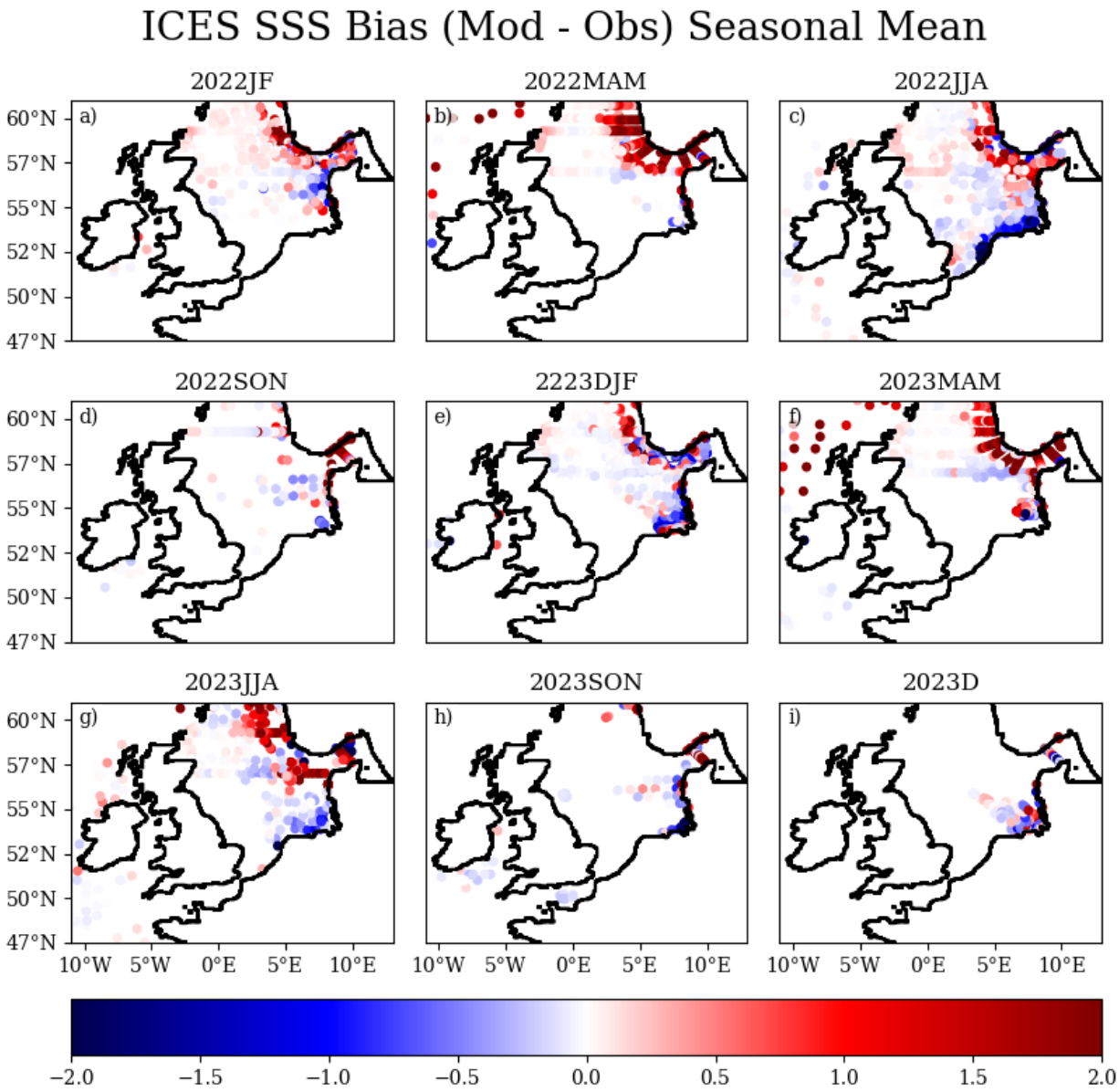


Figure 14: Seasonal model minus observation anomalies relative to ICES SSS. DJF, MAM, JJA and SON refer to winter, spring, summer and autumn. See Meterological seasons (section 0) for further clarification of the season definitions.

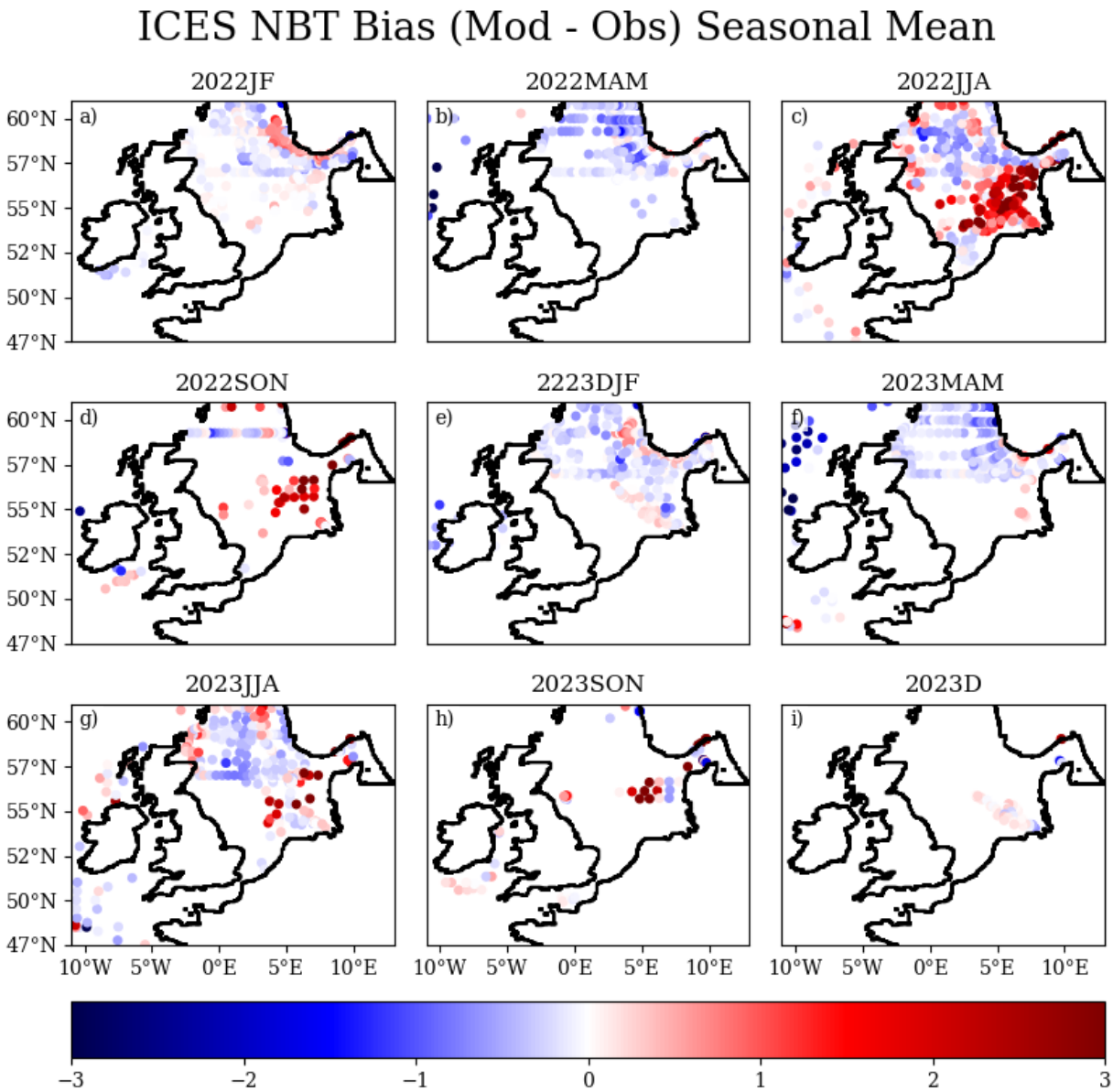


Figure 15: Seasonal model minus observation anomalies relative to ICES NBT. DJF, MAM, JJA and SON refer to winter, spring, summer and autumn. See Meterological seasons (section 0) for further clarification of the season definitions.

IV.4.3 Glider

The depth structure of hourly temperature and salinity is assessed against glider profiles from the HECLA (West of Scotland) and MOGLI (due East from Orkney) deployments, both available from BODC [8]. The glider data are separated into upward and downward casts and only the downward ones are used here. The model temperatures and salinities are matched up in time and space with the observations. Mixed

[8] https://platforms.bodc.ac.uk/deployment-catalogue/

Layer Depth (MLD) is assessed in section IV.5, but is used here to separate upper and lower layers. Statistics are estimated for temperature, and salinity above and below the mixed layer.

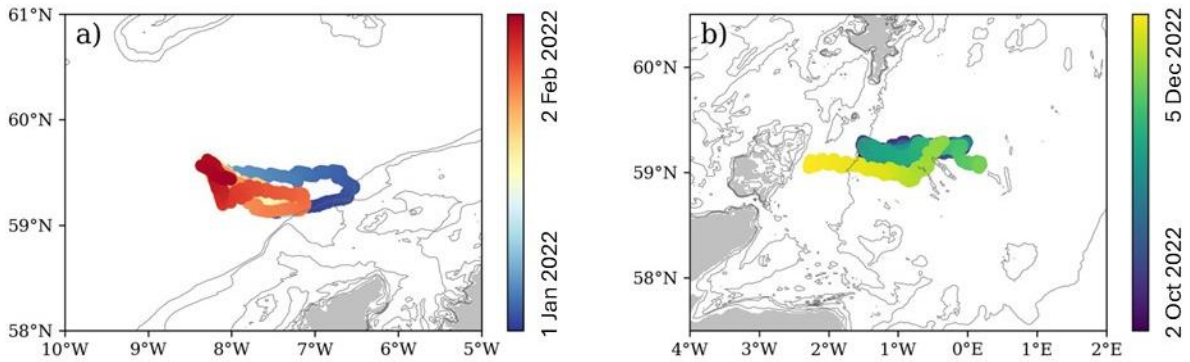


Figure 16: Glider tracks for the HECLA (a) and MOGLI (b) deployment.

Deployments tracks are shown in Figure 16. The HECLA track goes back and forth from the shelf break to a deeper point to the West between 1 January and 5 February 2022. The MOGLI track goes back and forth between the 100m isobath and a deeper point to the East between 2 October and 5 December 2022. This track crosses the region of a major part of the North Atlantic inflow into the North Sea.

The model generally represents the temperature depth variability well, with similar statistics for both deployments above the MLD (see Table 14). SST observations are assimilated into the mixed layer, while below the MLD a lot less observations are assimilated. As a result, both RMSD and bias are larger below the MLD with a general over-estimation of temperature along both tracks. For the MOGLI deployment the model strongly underestimates the temperature by a few degrees below the mixed layer. Figure 18 shows that this the bottom layer temperature anomalies are much smaller when the water is deeper, suggesting to the eastern end of the track.

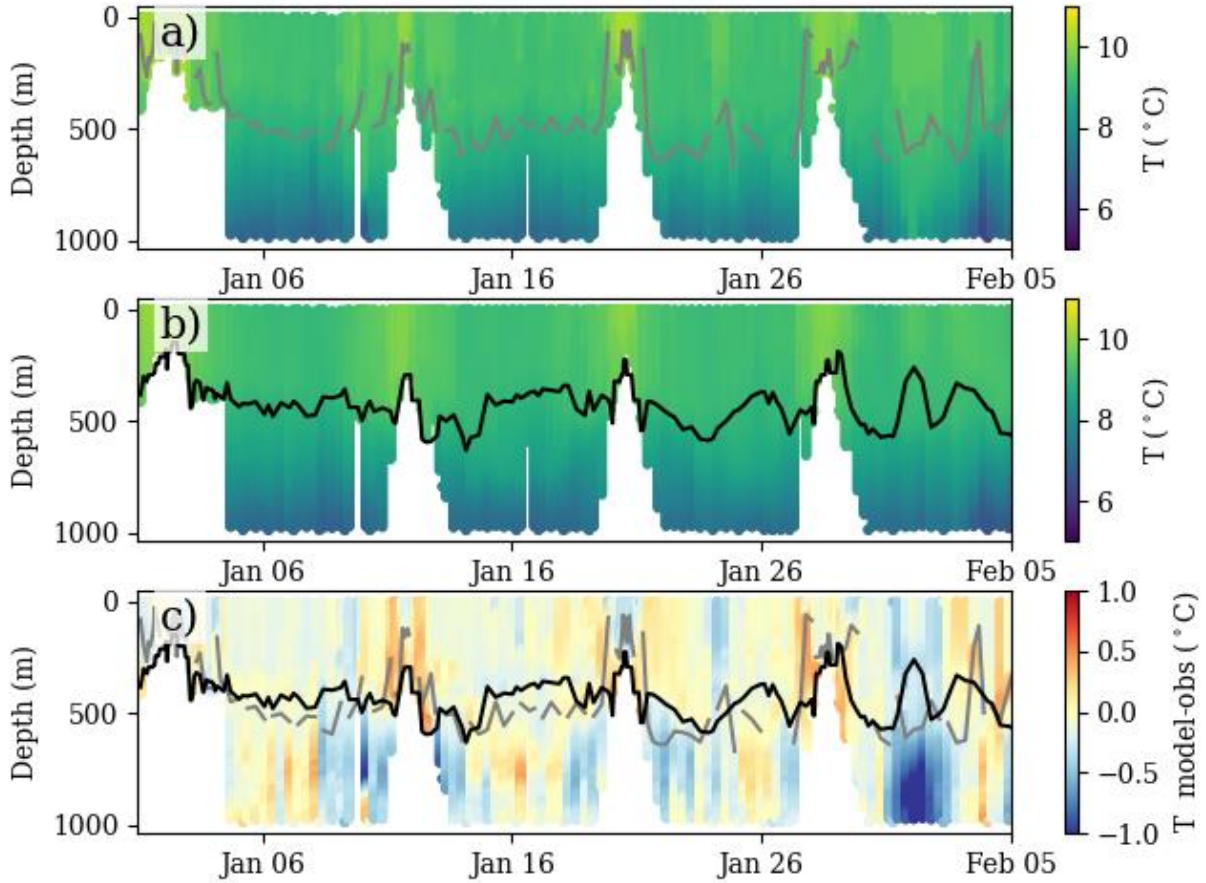


Figure 17: Temperature profile along the HECLA glider track from observations (a), from model (b) and difference between both (c). MLD from observations (grey line), and from the model (black line) are shown on the respective plots.

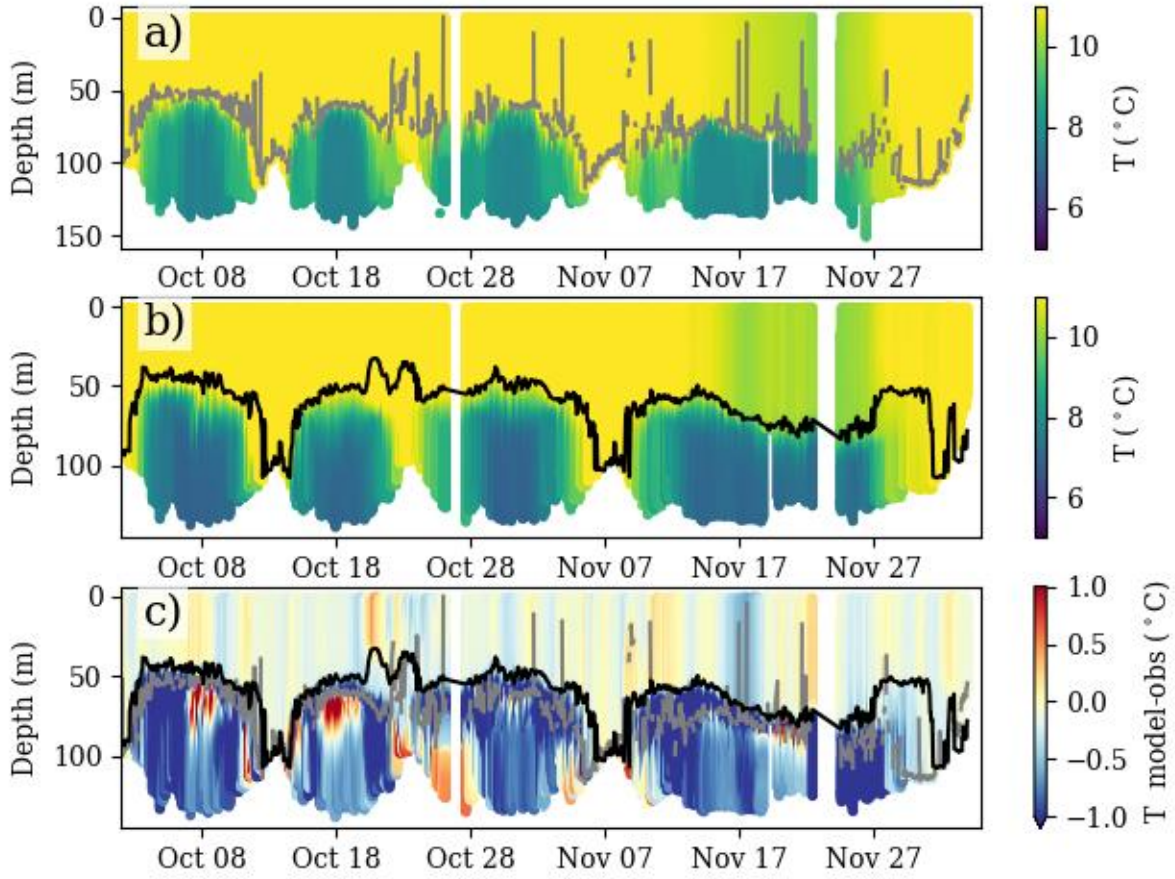


Figure 18: Temperature profile along the MOGLI glider track from observations (a), from model (b) and difference between both (c). MLD from observations (grey line), and from the model (black line) are shown on the respective plots.

The skill of the model for salinity is different between the two deployments. For the East of Scotland HECLA deployment, salinity differences between model and observations are small and evenly distributed above and below the MLD. For the MOGLI deployment, salinity is underestimated, and more strongly so above the MLD. At the end of the deployment, the observed salinities drop suddenly to very low values (nearly 30 psu), while the model shows a more gradual freshening. These observations appear unrealistic, and so summary statistics in Table 14 are repeated to exclude them.

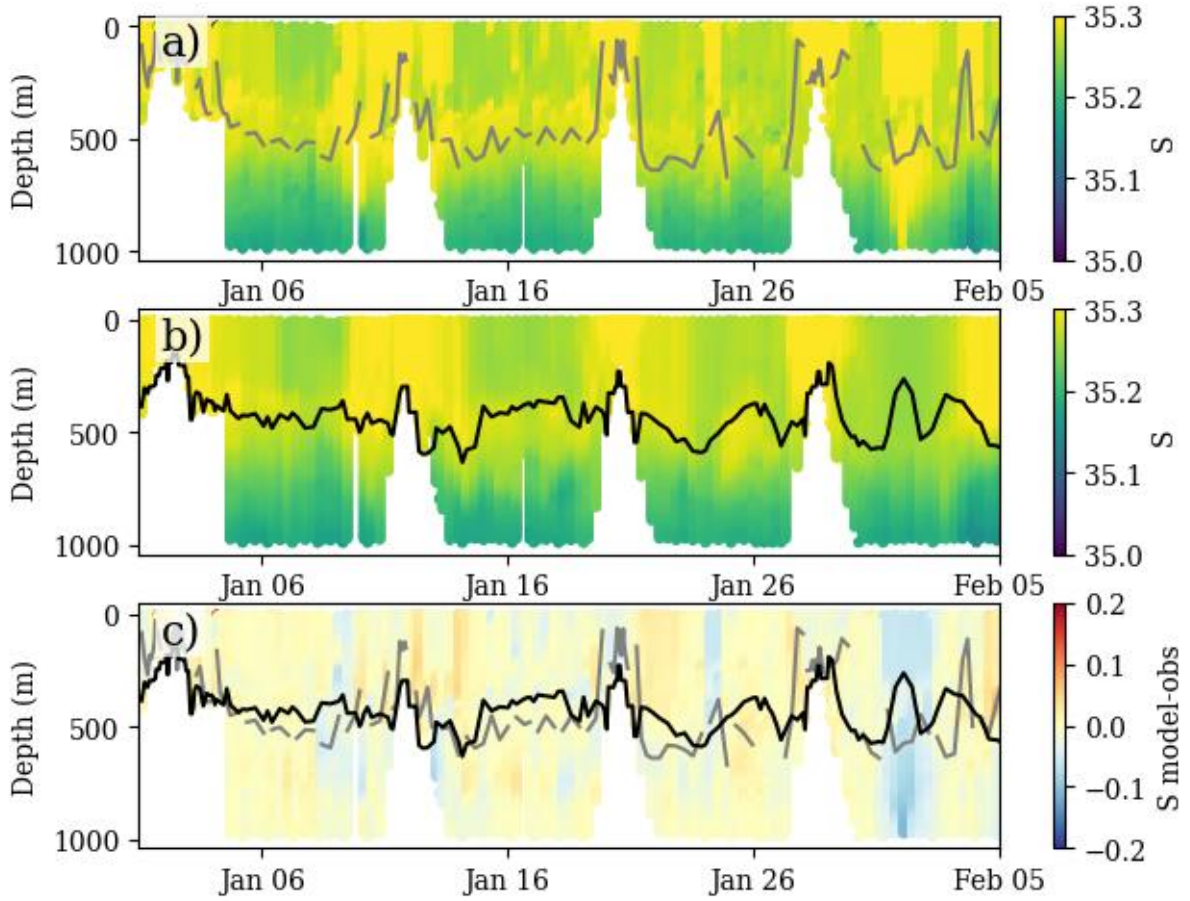


Figure 19: Salinity profile along the HECLA glider track from observations (a), from model (b) and difference between both (c). MLD from observations (grey line), and from the model (black line) are shown on the respective plots.

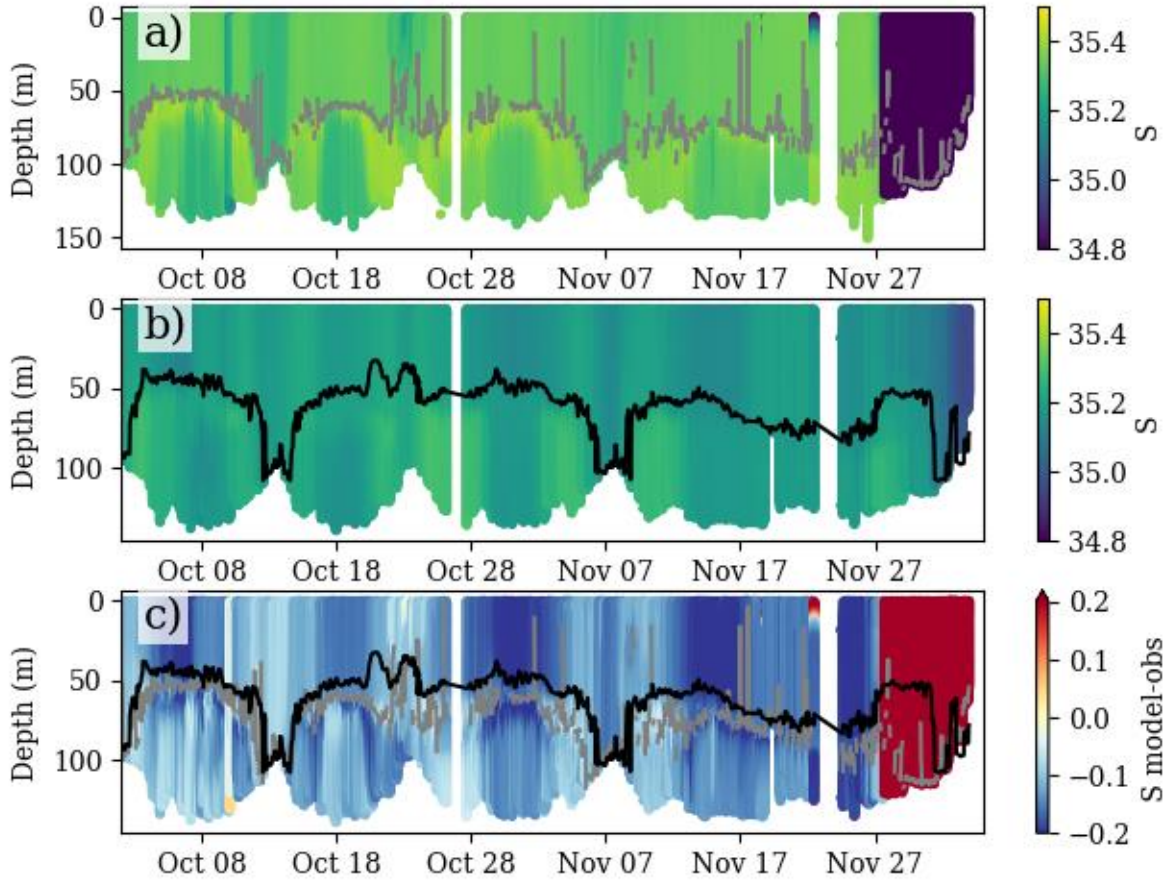


Figure 20: Salinity profile along the MOGLI glider track from observations (a), from model (b) and difference between both (c). MLD from observations (grey line), and from the model (black line) are shown on the respective plots. Observed salinity abruptly drops to value ~30 after 27 November. This results in difference between model and observations to nearly 5.

Table 14: Summary statistics for temperature and salinity above and below the MLD based on glider observations. The MOGLI salinity data was suspiciously fresh at the end of the deployment, so the statistics are repeated to exclude (include) these observations.

Deployment	**Metrics**	**Temperature** (°C) Above / Below	**Salinity (psu)** Above / Below	**Number of points** Above / Below
HECLA	Bias	-0.08 / -0.13	0.01 / 0.01	15514 / 9417
	RMSD	0.18 / 0.31	0.02 / 0.02	
MOGLI	Bias	-0.30 / -1.09	*0.45 / -0.09* -0.15 / -0.14	*87706 / 68492*
	RMSD	0.72 / 1.51	*1.70 / 0.50* 0.16 / 0.15	75738 /67226

IV.5 Mixed Layer Depth

Model MLD performance can only be assessed to a limited degree, due to a lack of available observations. We assess the modelled MLD against the two gliders, and find a good model-observation agreement in both the North Sea deployment, and outer Hebridean deployment, with relative biases of ~15%, relative RMSD <50% and that both have significantly correlated observed and modelled MLDs.

IV.5.1 Comparison with Gliders

Mixed Layer Depth (MLD) is estimated on the glider profiles using the de Boyer Montégut (2004) algorithm using a reference depth of 3m and a temperature of 0.02 °C. MLD calculated with the density criteria was very noisy as the glider salinity observations are also noisy, and so are not assessed.

Observed and modelled MLD are shown in Figure 17-Figure 20. MLD in both deployment' areas are very much controlled by temperature. Despite the larger salinity differences between model and observations for MOGLI, the MLD is better resolved for that deployment. This is possibly due to the strong gradient in temperature, which is well resolved, and the associated well defined MLD. For the HECLA deployment, bias and RMSD are larger, and correspond to much deeper MLD than for the MOGLI deployment. Although there are differences between observed and modelled MLD which increase with increasing MLD, the spatial and temporal pattern of the MLD is consistently reproduced by the model.

The HECLA and MOGLI absolute MLD biases of 53m and -10m correspond to relative biases of 14.5% and -17.09% (when considering the mean MLD in each region) while the RMSDs of 163 m and 20 m correspond to relative RMSDs of 46% and 34% respectively. The observed and modelled MLD time-series are significantly correlated with r^2 = 36% and r^2 = 53% respectively. However the time series are quite noisy, so may require a low pass filter prior to correlating to get a representative estimate.

Table 15: Summary statistics of MLD based on glider observations.

	Mean Obs. MLD (m)	Mean Model MLD (m)	Bias (model minus obs.; m)	RMSD	Relative Bias	Relative RMSD	Corr. (r^2) [r>0]	p	N
HECLA	328	386	53	163	14.85	45.66	0.36	~0.000	196
MOGLI	63	54	-10	20	-17.09	34.19	0.5329	~0.000	1333

IV.6 Sea Level Anomaly

The sea level of the NWS responds to different processes on different temporal and spatial scales. The tides of the NWS are very large and important, and so are specifically evaluated in section IV.2. The sea surface also responds to atmospheric conditions and oceanographic processes, which can change the mean sea level (compared to the long-term mean), when averaged over period of days – this is termed the sea level anomaly. Here we assess the performance of the model at simulating these sea level anomalies, against satellites. Satellite measurements of SLA are assimilated into the model, and so here we assess the model feedback files with the Obsstats package (see section III.2).

We show that AMM7 simulates the sea level anomalies very well, with mean biases of -0.56 cm (RMSD 7.13 cm). There are spatial patterns in both the bias and RMSD, with positive biases around Scotland and the Southern North Sea, and higher RMSDs along the European and Irish Sea coasts.

IV.6.1 Satellite sea level anomalies

AMM7 Sea Level Anomaly is assessed against the satellite SLA data used for the data assimilation (see Figure 5). The mean daily bias is very small ~-0.575 cm (RMSD = 7.011 cm, Figure 21). When calculated from all the observations in the domain, the bias is -0.56 cm, with an RMSD of 7.13cm (Table 16). The bias varies across the domain, with the model mean sea surface being slightly higher, on average, in the southern North Sea, and around Scotland (and down towards Wales), and lower in the open ocean region. This is in addition to the pattern of the mean dynamic topography, which show a similar on-shelf/off-shelf relationship. This pattern is reflected in the Table 16 with the Southern North Sea, Shetland shelf and Irish Sea regions positive biases (0.14 cm, 0.38 cm and 0.50 cm respectively) which are offset by the larger negative biases in the Atlantic regions (-1.01 cm and -0.51 cm for the southern and northern North Atlantic regions).

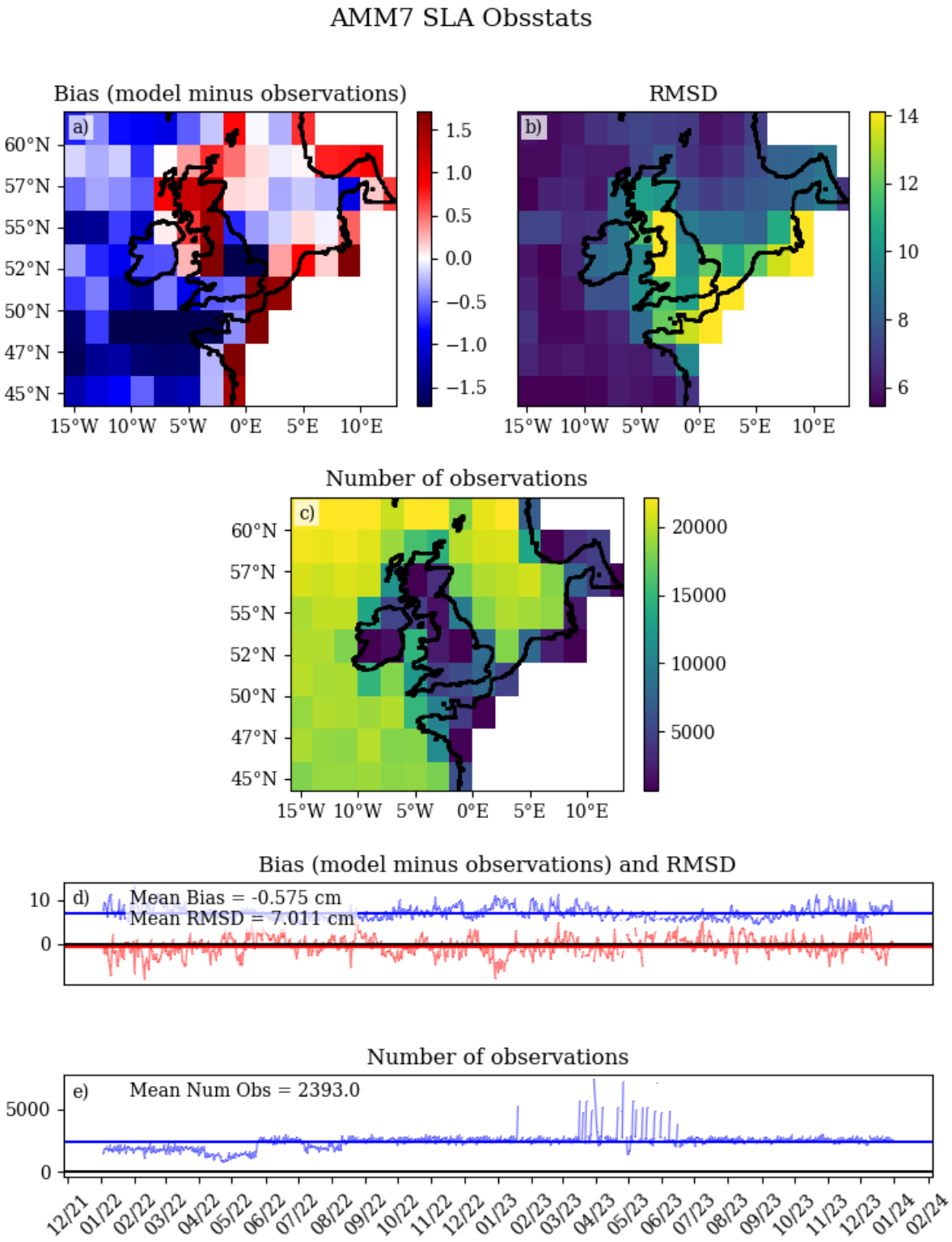


Figure 21: Model minus Observation daily SLA assessment against satellite observations. Maps of Bias, RMSD and Number of observations for 2022-2023 (a, b, c respectively) and time series of d) domain mean bias (blue) and RMSD (red) and e) number of observations per day.

Time series in Figure 21 provide daily statistics of all observations (i.e., model minus observation daily SLA against satellite observations) within the domain – the mean bias, RMSD and number of observations given in d) and e) are the average of these time series, thus they are not directly comparable to the values given in Table 16, which are calculated from all the observations within the Obsstat regions.

As expected in a tidally dominated area, on the continental shelf, the RMSD is higher than off-shelf, despite the constant presence of mesoscale eddies in the oceanic region adjacent to the NWS. The largest RMSD appears to be along the European continent, form the English Channel, Southern North Sea, and into the Skagerrak and Kattegat, as well as the Irish Sea, and this is reflected by the higher RMSDs in Table 16 for these regions (12.96 cm, 12.22 cm, 8.66 cm and 10.76 cm respectively), compared to a domain and NWS mean RMSDs of 7.13 and 8.4 cm (respectively).

The sea level responds to mechanisms such as steric expansion (higher in warmer fresher conditions), ocean dynamics (such as currents), and atmospheric conditions (such as storm surge events). The SLA biases and RMSDs also reflect differences in how these processes are represented in the model compared to reality. Some regions with the larger biases and RMSDs are important for certain mechanisms, which may explain these spatial patterns.

SLA data is now assimilated on the shelf (previously SLA assimilation is done only where the bathymetry is deeper than 700 m), and the SLA increments are calculated from detided model and observations (prior to being added back to the fully tidal model). This results in fewer observations being rejected, allowing a much greater number of observations to be assimilated

Table 16: SLA regional mean Obsstats summaries, calculated from all observations within the region, as defined in Figure 3d.

Region	Bias (cm)	RMSD (cm)	Number of Observations	Number of 2°x2° bins
NWS	-0.27	8.41	576641	35
Full Domain	-0.56	7.13	1608700	94
Southern North Sea	0.14	12.22	33873	3
Central North Sea	-0.17	9.07	85180	5
Northern North Sea	0.09	7.34	172122	9
English Channel	-0.40	12.96	9958	1
Skagerrak Kattegat	0.02	8.66	4491	1
Norwegian Trench	-0.00	6.81	35366	2
Shetland Shelf	0.38	6.38	39635	2
Irish Shelf	-0.26	6.89	102893	7
Irish Sea	0.50	10.76	14683	1
Celtic Sea	-1.37	7.32	118297	7
Armorican Shelf	-0.61	6.46	29167	2
NE Atlantic S	-1.01	5.98	486840	29
NE Atlantic N	-0.55	6.35	476195	25

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

Date	Change/Event description	System version	Other
Nov 2025	Entry into Service, replacing previous service (new Production Unit).	AMM7	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This modelling system (hereinafter referred to as the NWS-MFC system) is a new system similar to the one available prior to November 2023, which was also delivered by the UK Met Office (hereinafter referred as the Met Office system). The NWS-MFC system is the same system as the Met Office system, but the lateral boundary conditions are now provided by the Met Office. We now also assimilate SLA on shelf.

VII REFERENCES

Barbosa Aguiar, A., Bell, M. J., Blockley, E., Calvert, D., Crocker, R., Inverarity, G., King, R., Lea, D. J., Maksymczuk, J., Martin, M. J., Price, M. R., Siddorn, J., Smout-Day, K., Waters, J., and While, J.: The Met Office Forecast Ocean Assimilation Model (FOAM) using a 1/12-degree grid for global forecasts, Q. J. R. Meteorol. Soc., 150, 3827–3852, https://doi.org/10.1002/qj.4798, 2024.

Bloom, S. C., Takacs, L. L., Da Silva, A. M., and Ledvina, D.: Data Assimilation Using Incremental Analysis Updates, Mon. Weather Rev., 124, 1256–1271, https://doi.org/10.1175/1520-0493(1996)124%253C1256:DAUIAU%253E2.0.CO;2, 1996.

Butenschön, M., Clark, J., Aldridge, J. N., Allen, J. I., Artioli, Y., Blackford, J., Bruggeman, J., Cazenave, P., Ciavatta, S., Kay, S., Lessin, G., van Leeuwen, S., van der Molen, J., De Mora, L., Polimene, L., Sailley, S., Stephens, N., and Torres, R.: ERSEM 15.06: a generic model for marine biogeochemistry and the ecosystem dynamics of the lower trophic levels, Geosci. Model Dev., 9, 1293–1339, https://doi.org/10.5194/gmd-9-1293-2016, 2016.

De Boyer Montégut, C., Madec, G., Fischer, A. S., Lazar, A., and Iudicone, D.: Mixed layer depth over the global ocean: An examination of profile data and a profile-based climatology, J. Geophys. Res. Oceans, 109, 2004JC002378, https://doi.org/10.1029/2004JC002378, 2004.

Edwards, K. P., Barciela, R., and Butenschön, M.: Validation of the NEMO-ERSEM operational ecosystem model for the North West European Continental Shelf, Ocean Sci., 8, 983–1000, https://doi.org/10.5194/os-8-983-2012, 2012.

Flather, R. A.: A tidal model of the northwest European continental shelf, Mém. Société R. Sci. Liège, 10, 141–164, 1976.

Flather, R. A.: Results from a model of the north-east Atlantic relating to the Norwegian coastal current, in: The Norwegian Coastal Current, edited by: Sætre, R. and Mork, M., Bergen University, Bergen, Norway, 427–458, 1981.

Foreman, M. G. G.: Manual for tidal heights analysis and prediction, Institute of Ocean Sciences, Patricia Bay, Sidney, B.C., 1977.

Fowler, A. M., Skákala, J., and Ford, D.: Validating and improving the uncertainty assumptions for the assimilation of ocean-colour-derived chlorophyll *a* into a marine biogeochemistry model of the Northwest European Shelf Seas, Q. J. R. Meteorol. Soc., 149, 300–324, https://doi.org/10.1002/qj.4408, 2023.

Good, S., Fiedler, E., Mao, C., Martin, M. J., Maycock, A., Reid, R., Roberts-Jones, J., Searle, T., Waters, J., While, J., and Worsfold, M.: The Current Configuration of the OSTIA System for Operational Production of Foundation Sea Surface Temperature and Ice Concentration Analyses, Remote Sens., 12, 720, https://doi.org/10.3390/rs12040720, 2020.

Graham, J. A., O'Dea, E., Holt, J., Polton, J., Hewitt, H., Furner, R., Guihou, K., Brereton, A., Arnold, A., Wakelin, S., Castillo Sanchez, J. M., and Mayorga Adame, C. G.: AMM15: a new high-resolution NEMO configuration for operational simulation of the European north-west shelf, Geosci. Model Dev., 11, 681–696, https://doi.org/10.5194/gmd-11-681-2018, 2018.

Holt, J. T. and James, I. D.: An s coordinate density evolving model of the northwest European continental shelf - 1, Model description and density structure, J. Geophys. Res.-Oceans, 106, 14015–14034, 2001.

Howarth, M. J. and Pugh, D. T.: Chapter 4 Observations of Tides Over the Continental Shelf of North-West Europe, in: Elsevier Oceanography Series, vol. 35, Elsevier, 135–188, https://doi.org/10.1016/S0422-9894(08)70502-6, 1983.

Ingleby, B. and Huddleston, M.: Quality control of ocean temperature and salinity profiles - historical and real-time data, J. Mar. Syst., 65, 158–175, https://doi.org/10.1016/j.jmarsys.2005.11.019, 2007.

King, R. R., While, J., Martin, M. J., Lea, D. J., Lemieux-Dudon, B., Waters, J., and O'Dea, E.: Improving the initialisation of the Met Office operational shelf-seas model, Ocean Model., https://doi.org/10.1016/j.ocemod.2018.07.004, 2018.

Lea, D. J., Drecourt, J. -P., Haines, K., and Martin, M. J.: Ocean altimeter assimilation with observational- and model-bias correction, Q. J. R. Meteorol. Soc., 134, 1761–1774, https://doi.org/10.1002/qj.320, 2008.

Madec, G., Delecluse, P., Crépon, M., and Lott, F.: Large-Scale Preconditioning of Deep-Water Formation in the Northwestern Mediterranean Sea, J. Phys. Oceanogr., 26, 1393–1408, https://doi.org/10.1175/1520-0485(1996)026%253C1393:LSPODW%253E2.0.CO;2, 1996.

Madec, G., Bourdallé-Badie, R., Pierre-Antoine Bouttier, Bricaud, C., Bruciaferri, D., Calvert, D., Chanut, J., Clementi, E., Coward, A., Delrosso, D., Ethé, C., Flavoni, S., Graham, T., Harle, J., Iovino, D., Lea, D., Lévy, C., Lovato, T., Martin, N., Masson, S., Mocavero, S., Paul, J., Rousset, C., Storkey, D., Storto, A., and Vancoppenolle, M.: NEMO ocean engine, https://doi.org/10.5281/ZENODO.3248739, 2017.

Rio, M. H., Guinehut, S., and Larnicol, G.: New CNES-CLS09 global mean dynamic topography computed from the combination of GRACE data, altimetry, and in situ measurements, J. Geophys. Res. Oceans, 116, 2010JC006505, https://doi.org/10.1029/2010JC006505, 2011.

Siddorn, J. R. R. and Furner, R.: An analytical stretching function that combines the best attributes of geopotential and terrain-following vertical coordinates, Ocean Model., 66, 1–13, https://doi.org/10.1016/j.ocemod.2013.02.001, 2013.

Skákala, J., Ford, D., Brewin, R. J. W., McEwan, R., Kay, S., Taylor, B., De Mora, L., and Ciavatta, S.: The Assimilation of Phytoplankton Functional Types for Operational Forecasting in the Northwest European Shelf, J. Geophys. Res. Oceans, 123, 5230–5247, https://doi.org/10.1029/2018JC014153, 2018.

Tonani, M., Sykes, P., King, R. R., McConnell, N., Péquignet, A. C., O'Dea, E., Graham, J. A., Polton, J., and Siddorn, J.: The impact of a new high-resolution ocean model on the Met Office North-West European Shelf forecasting system, Ocean Sci., https://doi.org/10.5194/os-15-1133-2019, 2019.

Umlauf, L. and Burchard, H.: A generic length-scale equation for geophysical turbulence models, J. Mar. Res., 61, 235–265, https://doi.org/10.1357/002224003322005087, 2003.

Vörösmarty, C. J., Fekete, B. M., Meybeck, M., and Lammers, R. B.: Geomorphometric attributes of the global system of rivers at 30-minute spatial resolution, J. Hydrol., 237, 17–39, 2000.

Wakelin, S. L., Holt, J., Blackford, J., Allen, I., Butenschön, M., and Artioli, Y.: Modeling the carbon fluxes of the northwest European continental shelf: Validation and budgets, J. Geophys. Res. Oceans, 117, C05020, https://doi.org/DOI:%252010.1029/2011JC007402, 2012.

While, J. and Martin, M. J.: Variational bias correction of satellite sea-surface temperature data incorporating observations of the bias, Q. J. R. Meteorol. Soc., 145, 2733–2754, https://doi.org/10.1002/qj.3590, 2019.

Willett, M. R., Brooks, M., Bushell, A., Earnshaw, P., Smith, S., Tomassini, L., Best, M., Boutle, I., Brooke, J., Edwards, J. M., Furtado, K., Hardacre, C., Hartley, A. J., Hewitt, A., Johnson, B., Lock, A., Malcolm, A., Mulcahy, J., Müller, E., Rumbold, H., Rooney, G. G., Sellar, A., Ujiie, M., Van Niekerk, A., Wiltshire, A., and Whitall, M.: The Met Office Unified Model Global Atmosphere 8.0 and JULES Global Land 9.0 configurations, https://doi.org/10.5194/egusphere-2025-1829, 28 May 2025.

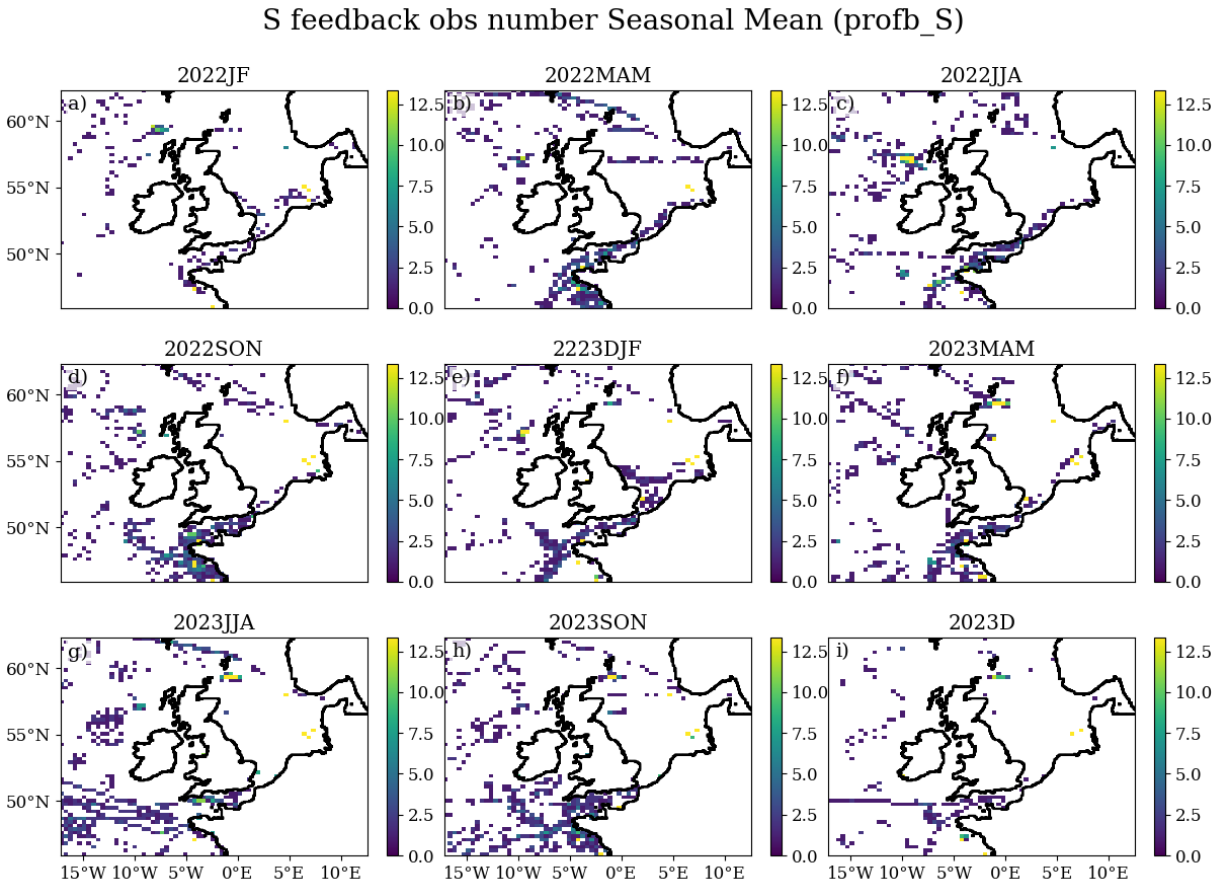
Xavier, P., Willett, M., Graham, T., Earnshaw, P., Copsey, D., Marzin, C., Sellar, A., Ackerley, D., Alves, O., Blockley, E., Bodas-Salcedo, A., Bushell, A., Butchart, N., Calvert, D., Cho, J.-A., De Burgh-Day, C., Edwards, J., Furtado, K., Field, P., Guiavarch, C., Hardy, S., Harris, C., Heywood, K., Heming, J., Hendon, H., Hewitt, H., Hyder, P., Hyun, Y.-K., Hyun, S.-H., Ineson, S., Jones, R., Kim, J., Kim, K.-Y., Klingaman, N., Levine, R., Lee, S.-M., Lekakou, K., Lock, A., Martin, G., Mathiot, P., Megann, A., Meijers, A., Moon, J.-Y., Morgenstern, O., North, R., Nurser, G., Park, Y.-H., Regayre, L., Roberts, M., Rodriguez, J., Ridley, J., Rawlins, R., Sinha, B., Shin, B., Semple, A., Storkey, D., Stephens, D., Shim, T., Tomassini, L., Tsushima, Y., Titley, H., Tennant, W., Varma, V., Vellinga, M., Weedon, G., Williams, K., Yang, Y.-M., Zhao, M., Zhou, X., Zhu, H., Met Office, Exeter, UK, The Bureau of Meteorology, Melbourne, Australia, National Institute of Meteorological Sciences, Korea Meteorological Administration, Jeju, Korea, National Centre for Atmospheric Science (NCAS), University of Reading, Reading, United Kingdom, NIWA - The National Institute of Water and Atmospheric Research, New Zealand, Centre for Environmental Modelling and Computation, University of Leeds, Leeds, United Kingdom, School of Earth and Environment, University of Leeds, Leeds, United Kingdom, National Oceanography Centre, Southampton, United Kingdom, and Joint Marine Modelling Programme (JMMP): Assessment of the Met Office Global Coupled model version 4 (GC4) configurations., Met Office, https://doi.org/10.62998/uzui3766, 2024.

Young, E. F. and Holt, J. T.: Prediction and analysis of long-term variability of temperature and salinity in the Irish Sea, J. Geophys. Res.-Oceans, 112, https://doi.org/C01008%252010.1029/2005jc003386, 2007.

Zalesak, S. T.: Fully multidimensional flux-corrected transport algorithms for fluids, J. Comput. Phys., 31, 335–362, https://doi.org/10.1016/0021-9991(79)90051-2, 1979.

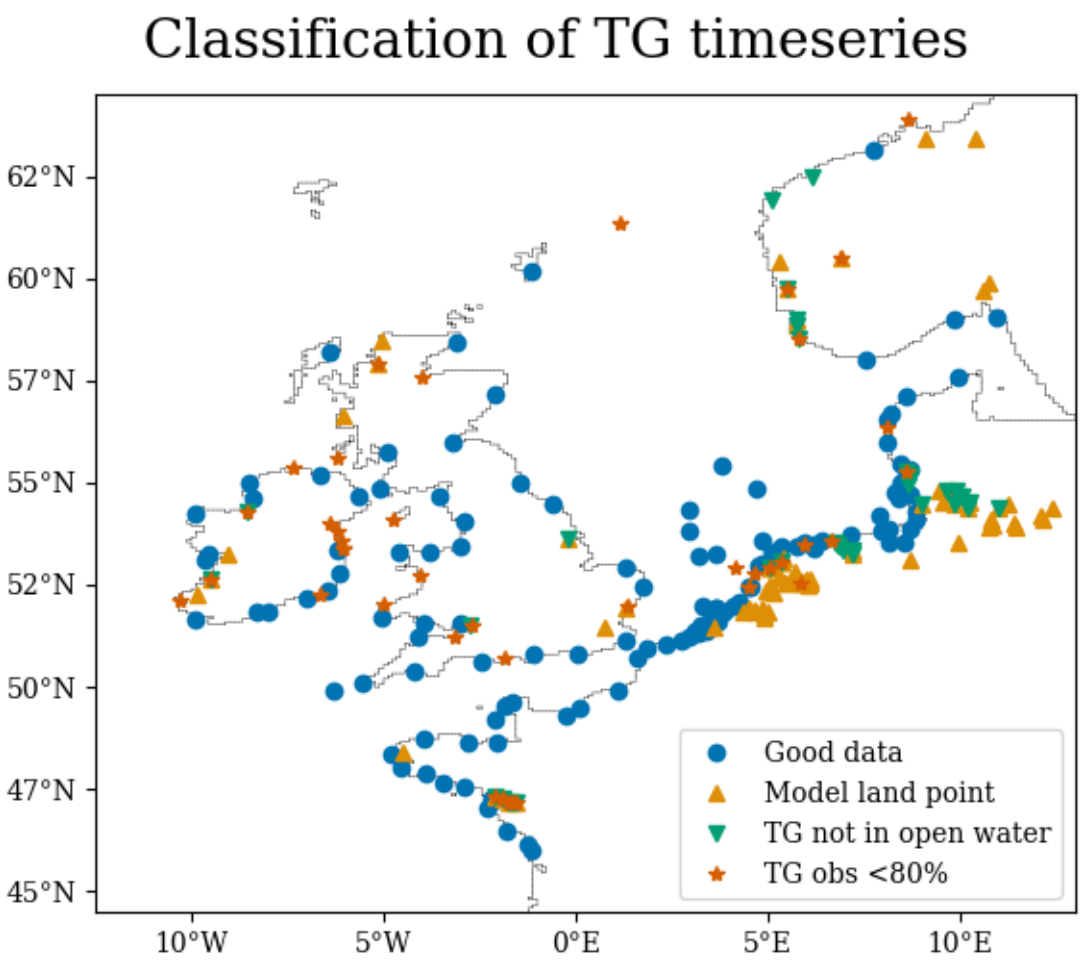
VIII APPENDIX

VIII.1 Additional Figures

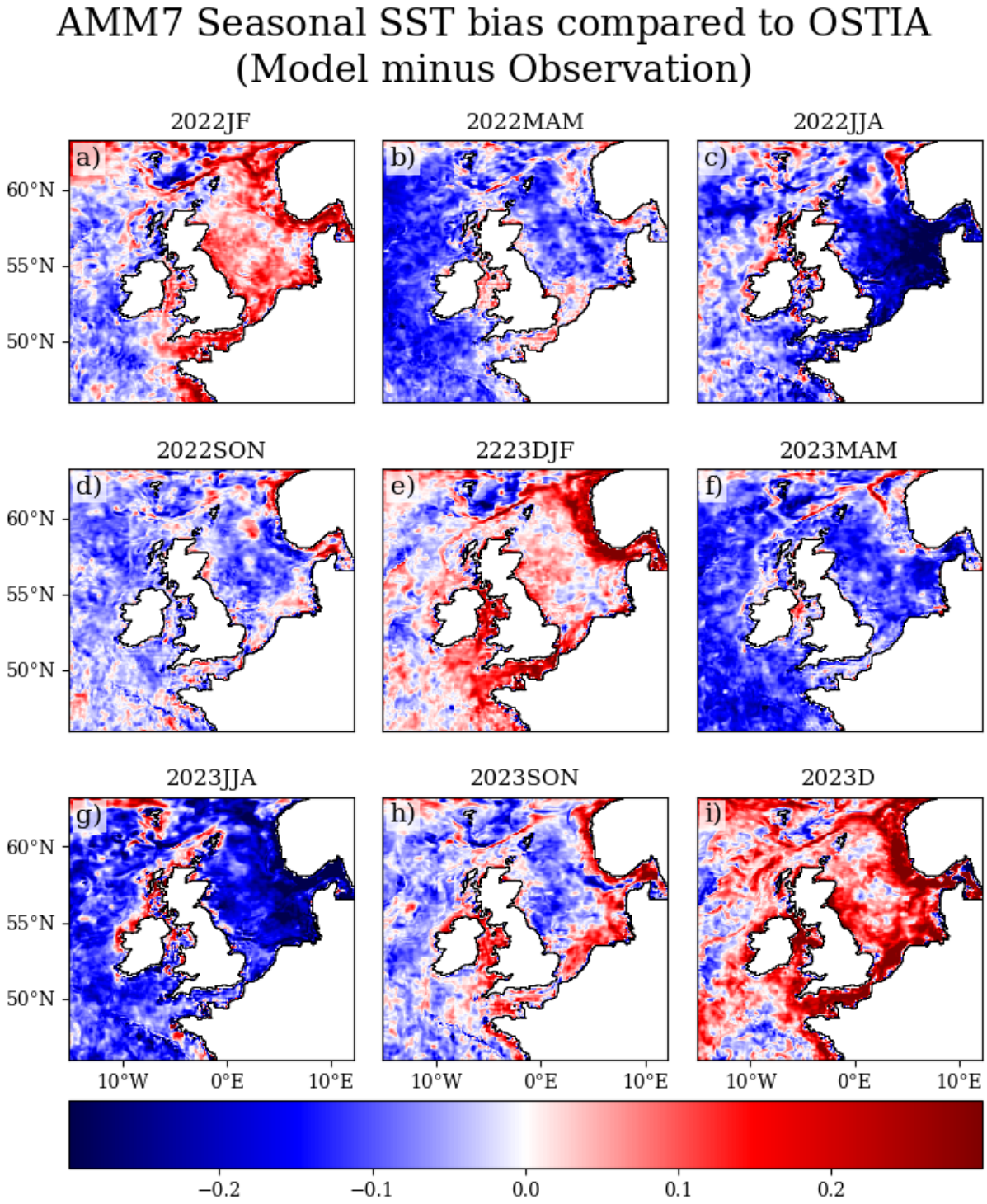


Appendix Figure 1: Total number of salinity profiles in 20 grid box bins for 2022-2023 presented by season. DJF, MAM, JJA and SON refer to winter, spring, summer and autumn. See

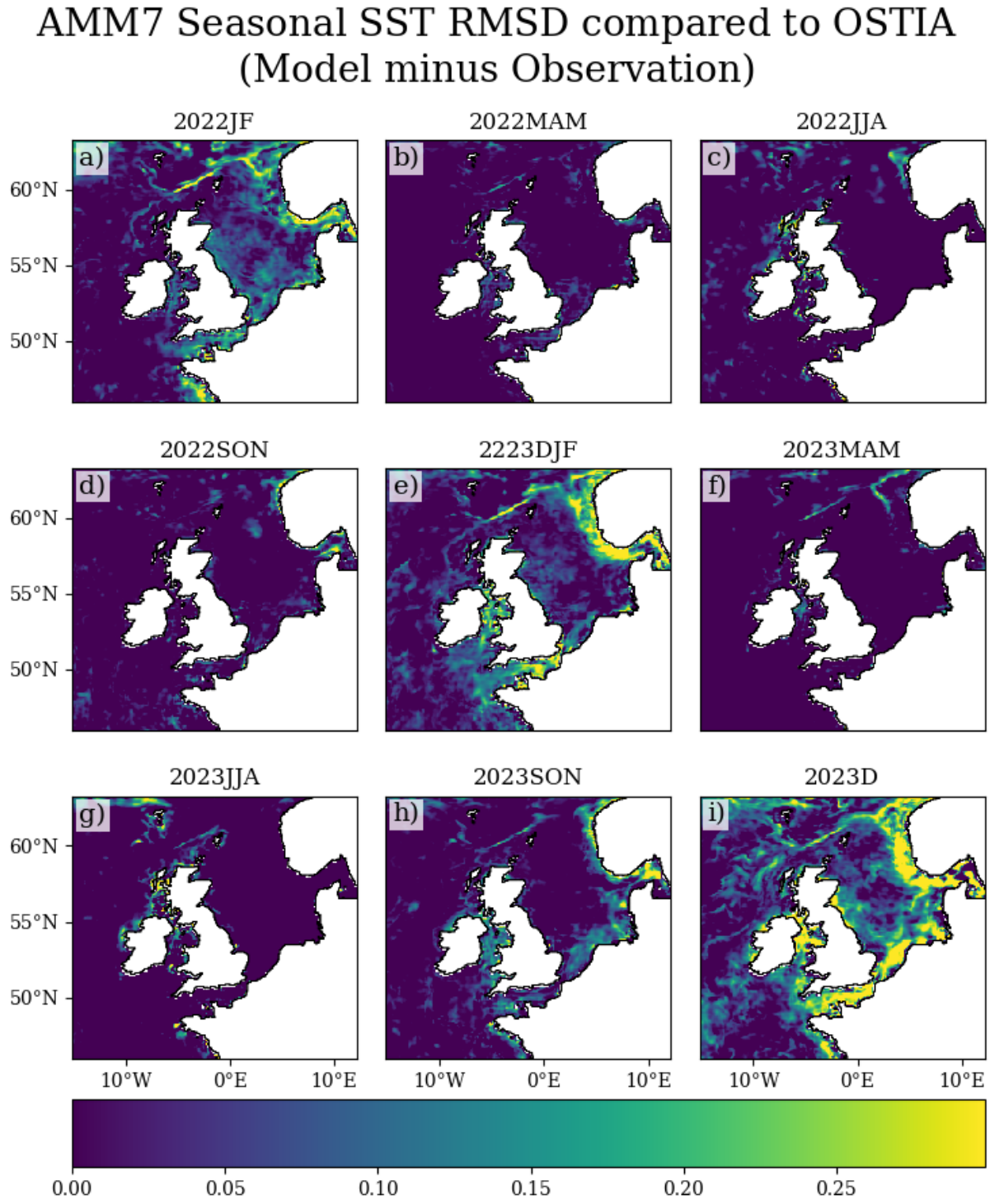
Meteorological seasons (section 0) for further clarification of the season definitions. See Section IV.1.



Appendix Figure 2: Classification of Copernicus tide gauges used in tidal analysis. This figure show the location of the tide gauges records downloaded from the Copernicus Marine INS-TAC. Some data was rejected as it was considered a land point on the model domain, not considered to be in open water (i.e.in an estuary, the Wadden Sea, the Belt Seas) or because the data coverage was less than 80% for the 2-year trial period (2022-2023; ★). See Section IV.2.1 for details.



Appendix Figure 3: Model foundation temperature (T_{4m}, at 4m depth) minus OSTIA bias for seasonal means (positive mean error indicates a warm model bias). DJF, MAM, JJA and SON refer to winter, spring, summer and autumn. See Meterological seasons (section 0) for further clarification of the season definitions. See Section IV.4.2.



Appendix Figure 4: Model foundation temperature (T_{4m}, at 4m depth) minus OSTIA RMSD for seasonal means. DJF, MAM, JJA and SON refer to winter, spring, summer and autumn. See Meterological seasons (section 0) for further clarification of the season definitions. See Section IV.4.2.

VIII.2 Additional Tables

Appendix Table 1: Details of the NEMO ocean model configuration.

NWSHELF_ANALYSISFORECAST		**AMM7 PHY_LR_004_001**	**AMM15 PHY_004_013**
Model	Bathymetry	GEBCO corrected by NOOS partners	EMODnet 2015 – see footnote 2 on page 12 for more information.
	Horizontal resolution	7 km	1.5 km
	Model Grid 40°N- 65°N 20°W – 13°E Regular grid	40°N-65°N 20°W-13°E Regular grid	~ 45°N - 63°N ~ 20°W - 13°E The grid has a rotated pole, chosen so that the grid-equator runs through the domain to reduce the distortion of cells with increasing latitude. While the rotated latitude is constant, the longitudinal grid steps range from ~1.47 km to 1.5 km.
	Timestep	60 s (~3.5 s barotropic sub-timestep)	300 s (10 s barotropic sub-timestep)
	Penetrative radiation	1-band shortwave radiation light attenuation (as used in POLCOMS, Holt and James, 2001)	NEMO tri-band Red-Blue-Green (RGB)
	Bottom friction	Log layer. Minimum drag coefficient 1.0×10^{-3}	Log layer. Minimum drag coefficient 2.5×10^{-3}
	Momentum diffusion	bi-Laplacian on model levels -1×10^{10} m^4/s	bi-Laplacian on model levels 6×10^{7} m^4/s
	Tracer diffusion	Laplacian on geopotential levels 50 m^2/s.	bi-Laplacian on model levels -1×10^{5} m^4/s

Appendix Table 2: Details of the surface and lateral boundary conditions.

NWSHELF_ANALYSISFORECAST		**AMM7 PHY_LR_004_001 and AMM15 PHY_004_013**
Forcing	Surface forcing	Met Office Global Unified Model (MetUM) Atmospheric model NWP analysis and forecast fields, applied as direct fluxes.
	Surface forcing resolution	Horizontal grid: ~10 km (2560 x 1920 grid points) Frequency: 3 hourly mean fluxes of long and short-wave radiation, moisture, 3 hourly mean air surface temperature but hourly 10 m winds and surface pressure
	Lateral boundaries	Met Office global ocean 12km model ORCA12, daily mean 3D Temperature, Salinity and barotropic currents, and sea surface height
		AMM7 and AMM15 have Atlantic and Baltic boundaries in a different geographical location.

Appendix Table 3: Details of the river boundary conditions.

NWSHELF_ANALYSISFORECAST		**AMM7 PHY_LR_004_001**	**AMM15 PHY_004_013**
Forcing	River run-off	Daily climatology of gauge data averaged for 1950–2005. Climatology of daily discharge data for 279 rivers from the Global River Discharge Data Base (Vörösmarty et al., 2000) and from data prepared by the Centre for Ecology and Hydrology as used by (Young and Holt, 2007).	Daily climatology of gauge data averaged for 1980–2014. UK data were processed from raw data provided by the Environment Agency, the Scottish Environment Protection Agency, the Rivers Agency (Norther Ireland), and the National River Flow Archive (personal communication by Sonja M. van Leeuwen, CEFAS, 2016). For major rivers that were missing from this data set (e.g. along the French and Norwegian coast), data have been provided by the same climatology used by AMM7 (Vörösmarty et al., 2000; Young and Holt, 2007).

Appendix Table 4: Details of the NEMOVAR data assimilation scheme.

NWSHELF_ANALYSISFORECAST		**AMM7** **PHY_LR_004_001**	**AMM15** **PHY_004_013**
Data Assimilation	NEMOVAR version	V3	V4
	SST bias correction scheme:	Offline observation-of-bias scheme. Reference dataset: in-situ.	Variational scheme in addition to observations-of-bias. Reference datasets: in-situ (drifters only) and VIIRS satellite data.
	Correlation operator short scale: 3-times grid scale	~20 km	~5 km
	Data Assimilated	SST (*in situ*, and satellite), temperature and salinity profiles, glider data, sea level anomaly data, ocean colour chlorophyll	SST (*in situ*, and satellite), temperature and salinity profiles, glider data, sea level anomaly data

Appendix Table 5: Details of the ERSEM biogeochemistry model.

NWSHELF_ANALYSISFORECAST		**AMM7 PHY_LR_004_001**
BGC model	Model and version	ERSEM 22.11, https://github.com/pmlmodelling/ersem/releases/tag/22.11
	Description	Highly complex marine biogeochemistry and ecosystem model with pelagic and benthic components; ~50 state variables
	Pelagic	• Carbonate system (based on DIC and alkalinity); • nutrients (nitrogen, ammonium, silicate, phosphate, iron – not used in AMM7); • 4 phytoplankton functional types (picophytoplankton (< 2 µm), nanophytoplankton (2–20 µm), microphytoplankton (> 20 µm), and diatoms); • 3 zooplankton types (microzooplankton, heterotrophic flagellates and mesozooplankton); • microbes; dissolved and particulate organic matter; On NEMO grid
	Benthic	• Bacteria (aerobic and anaerobic); • benthic predators; • dissolved organic matter; • particulate detritus; • buried refractory matter; vertically integrated for three model layers
	Other features	• Fully dynamic stoichiometry for components (e.g. in phytoplankton: carbon, nitrate, phosphate(, silicate), chlorophyll); forced by physics (not two-way coupled) with multiple dependencies on physics (e.g. temperature-dependent metabolism, temperature and wind dependent air-sea flux of carbon dioxide and oxygen)
	Reference	(Butenschön et al., 2016; Edwards et al., 2012; Fowler et al., 2023; Skákala et al., 2018; Tonani et al., 2019).

Appendix Table 6: Details of key dates for the NWS-MFC consortia.

NWSHELF_ANALYSISFORECAST		**AMM7 PHY_LR_004_001**	**AMM15 PHY_004_013**
Service	Entry Into Service.	19/04/2017	18/11/2018
	Exited Service.	1/11/2022	1/11/2022
	Re-entered Into Service.	1/1/2025	1/1/2025
	Began redelivering with double dissemination.	28/10/2025	28/10/2025
	Became main for NWS-MFC NRTs	25/11/2025	25/11/2025

Appendix Table 7: Summary statistics of the tidal harmonic analysis, based on 146 tide gauges from the Copernicus INS TAC tide gauge product included in the analysis.

	Median (cm)	Mean (10th -90th percentiles; cm)	RMSD (cm)	Median (°)	Mean (10th -90th percentiles; °)	RMSD (°)
M_2	-3.40	-1.61 (-17.35 - 18.48)	12.19	-5.41	-4.63 (-18.87 - 12.06)	11.49
S_2	-0.13	-0.16 (-5.21 - 6.13)	3.69	-6.46	-3.64 (-17.68 - 26.47)	14.00
N_2	-0.58	-0.45 (-3.47 - 3.07)	2.01	-5.98	-4.94 (-18.17 - 14.40)	10.67
K_1	0.84	0.82 (-0.59 - 2.17)	1.18	2.73	2.35 (-14.97 - 16.98)	13.79
O_1	-0.94	-0.89 (-2.07 - 0.65)	1.20	-0.42	-1.95 (-9.09 - 14.90)	30.94
M_4	0.09	-0.04 (-8.94 - 8.60)	4.62	-40.73	-38.66 (-113.07 - 34.82)	61.75
Q_1	0.44	0.46 (-0.11 - 1.07)	0.58	-17.41	-14.56 (-25.95 - 0.77)	20.37

Appendix Table 8: Statistics of SST, NBT and SSS compared to ICES for the North Sea.

North Sea	Mean (Standard deviation)	Median (10th-90th percentile	RMSD	Number of Observation
SST				
2022 JF	-0.08°C (0.34°C)	-0.07°C (-0.42°C-0.22°C)	0.35°C	272
2022 MAM	-0.23°C (0.43°C)	-0.21°C (-0.54°C-0.12°C)	0.49°C	240
2022 JJA	-0.22°C (0.50°C)	-0.20°C (-0.67°C-0.23°C)	0.55°C	485
2022 SON	0.10°C (0.63°C)	0.00°C (-0.42°C-0.62°C)	0.64°C	116
22/23 DJF	-0.10°C (0.46°C)	-0.14°C (-0.45°C-0.23°C)	0.47°C	325
2023 MAM	-0.12°C (0.42°C)	-0.13°C (-0.49°C-0.31°C)	0.44°C	295
2023 JJA	-0.19°C (0.67°C)	-0.15°C (-0.60°C-0.29°C)	0.70°C	285
2023 SON	0.14°C (1.11°C)	-0.13°C (-0.53°C-0.55°C)	1.12°C	68
2023 D	0.06°C (0.37°C)	0.06°C (-0.35°C-0.61°C)	0.38°C	72
SSS				
2022 JF	0.20psu (1.00psu)	0.03psu (-0.16psu-0.42psu)	1.02psu	255
2022 MAM	0.48psu (1.19psu)	0.07psu (-0.11psu-1.51psu)	1.28psu	220
2022 JJA	0.57psu (3.55psu)	-0.01psu (-0.58psu-0.76psu)	3.59psu	463
2022 SON	1.19psu (4.84psu)	-0.04psu (-0.59psu-0.81psu)	4.98psu	99
22/23 DJF	0.33psu (3.41psu)	-0.02psu (-0.52psu-0.28psu)	3.42psu	320
2023 MAM	0.62psu (3.31psu)	0.04psu (-0.29psu-0.96psu)	3.37psu	295
2023 JJA	1.03psu (4.24psu)	0.01psu (-0.40psu-1.78psu)	4.36psu	285
2023 SON	2.97psu (9.20psu)	-0.07psu (-0.85psu-25.45psu)	9.67psu	68
2023 D	0.44psu (3.66psu)	-0.02psu (-0.63psu-0.88psu)	3.69psu	72
NBT				
2022 JF	-0.10°C (0.23°C)	-0.08°C (-0.41°C-0.12°C)	0.25°C	190
2022 MAM	-0.28°C (0.45°C)	-0.25°C (-0.67°C-0.02°C)	0.53°C	147
2022 JJA	0.66°C (1.23°C)	0.34°C (-0.59°C-2.33°C)	1.39°C	311
2022 SON	0.77°C (1.19°C)	0.52°C (-0.48°C-2.22°C)	1.42°C	60
22/23 DJF	-0.15°C (0.25°C)	-0.14°C (-0.43°C-0.17°C)	0.29°C	199
2023 MAM	-0.20°C (0.24°C)	-0.19°C (-0.50°C-0.06°C)	0.31°C	169
2023 JJA	0.08°C (0.80°C)	-0.10°C (-0.55°C-0.77°C)	0.81°C	219
2023 SON	1.11°C (1.88°C)	0.31°C (-0.48°C-4.40°C)	2.18°C	26
2023 D	0.03°C (0.23°C)	0.08°C (-0.32°C-0.22°C)	0.23°C	35
NBS				
2022 JF	0.00psu (0.12psu)	0.02psu (-0.14psu-0.11psu)	0.12psu	190
2022 MAM	0.00psu (0.12psu)	0.03psu (-0.15psu-0.09psu)	0.12psu	147
2022 JJA	-0.05psu (0.19psu)	-0.04psu (-0.24psu-0.13psu)	0.20psu	311
2022 SON	-0.16psu (0.31psu)	-0.04psu (-0.58psu-0.11psu)	0.35psu	60
22/23 DJF	-0.07psu (0.20psu)	-0.02psu (-0.16psu-0.05psu)	0.21psu	199
2023 MAM	-0.09psu (0.14psu)	-0.04psu (-0.29psu-0.04psu)	0.16psu	169
2023 JJA	-0.13psu (0.17psu)	-0.06psu (-0.38psu-0.02psu)	0.22psu	219
2023 SON	-0.11psu (0.22psu)	-0.05psu (-0.36psu-0.06psu)	0.24psu	26
2023 D	-0.14psu (0.51psu)	-0.04psu (-0.59psu-0.15psu)	0.53psu	35